

APPENDIX C

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B or equivalent) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5. If the Power of Attorney by Applicant form is not accompanied by this transmittal form or an equivalent, the Power of Attorney will not be recognized in the application.

Application Number	Unassigned
Filing Date	Herewith
First Named Inventor	Nicolas Breil
Title	SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS
Art Unit	Unassigned
Examiner Name	Unassigned
Attorney Docket Number	FIS920130015US1(29690)

SIGNATURE of Applicant or Patent Practitioner

Signature	/Leslie S. Szivos/	Date	08-29-2013
Name	Leslie S. Szivos, Ph.D.	Telephone	(516) 742-4343
Registration Number	39394		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications.



*Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the above-identified patent application.

- ☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent):

45600

OR

- ☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent):

Name Registration	Number		Name Registration	Number

Please recognize or change the correspondence address for the above-identified patent to:

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OR

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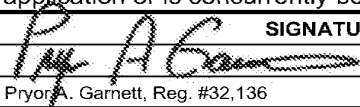
OR

☐ Firm or Individual Name				
Address				
City		State		Zip
Country				
Telephone		Email		

I am the Applicant:

- ☐ Inventor or Joint Inventor
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign
- ☐ Person Who Otherwise Shows Sufficient *Proprietary* Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document)

SIGNATURE of Applicant for Patent

Signature		Date	September 10, 2012
Name	Pryor A. Garnett, Reg. #32,136	Telephone	503-578-5775
Title and Company	Senior Counsel, International Business Machines Corporation		

NOTE: Signature - This form must be signed by the applicant or applicant's representative in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms for more than one signature, see below *.

- ☒ *Total of 1 (one) forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Nicolas Breil	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920130015US1(29690)	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	6063681		2000-05-16	Son	
	2	7737032		2010-06-15	Chan et al.	
	3	6815235		2004-11-09	Markle	
	4	7023055		2006-04-04	leong et al.	
	5	7329923		2008-02-12	Doris et al.	
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	1	20110287318		2011-11-24	Loveness et al.	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Nicolas Breil	
Art Unit		
Examiner Name		
Attorney Docket Number	FIS920130015US1(29690)	

2	20120175682		2012-07-12	Hagleitner et al.	
3	20120112292		2012-05-10	Lavoie et al.	
4	20100155954		2010-06-24	Mukherjee et al.	
5	20070284693		2007-12-13	Kim et al.	
6	20050116290		2005-06-02	de Souza et al.	

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Nicolas Breil
Art Unit	
Examiner Name	
Attorney Docket Number	FIS920130015US1(29690)

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Nicolas Breil
Art Unit	
Examiner Name	
Attorney Docket Number	FIS920130015US1(29690)

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Catherine Ivers/	Date (YYYY-MM-DD)	
Name/Print	Catherine Ivers	Registration Number	59565

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

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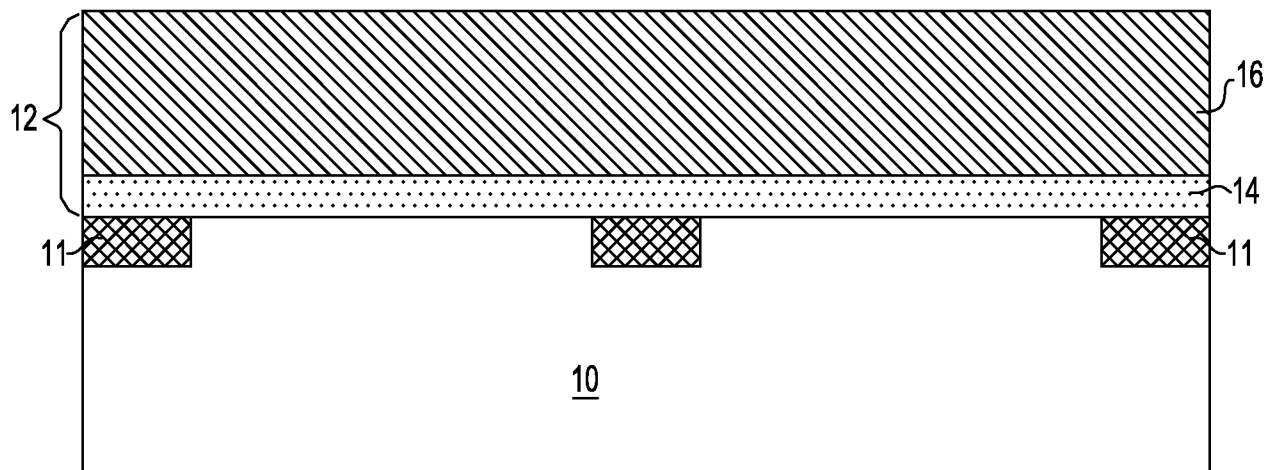


FIG. 1

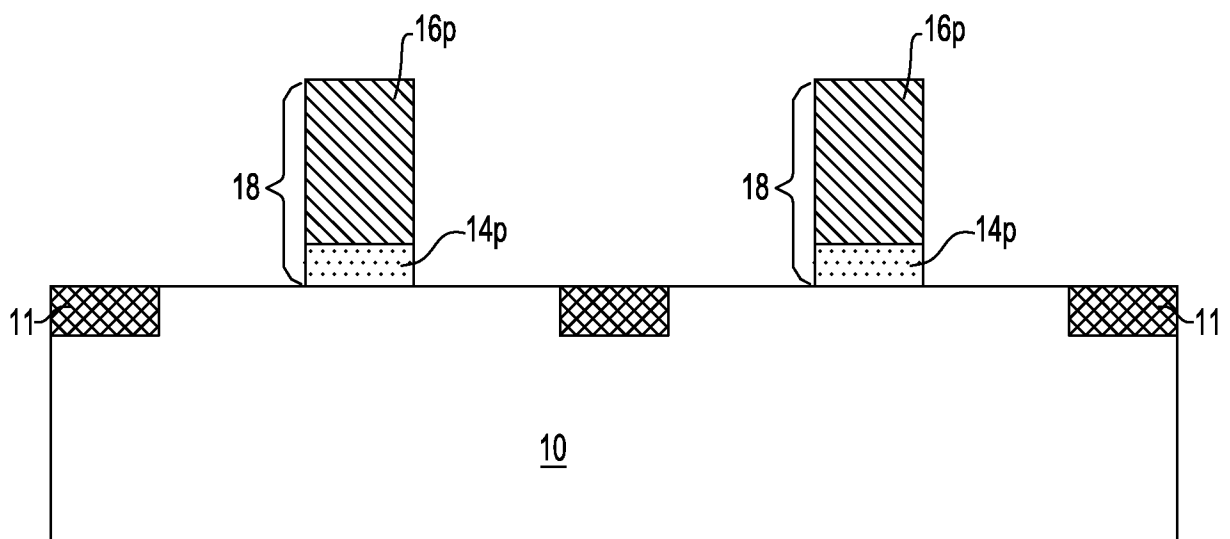


FIG. 2

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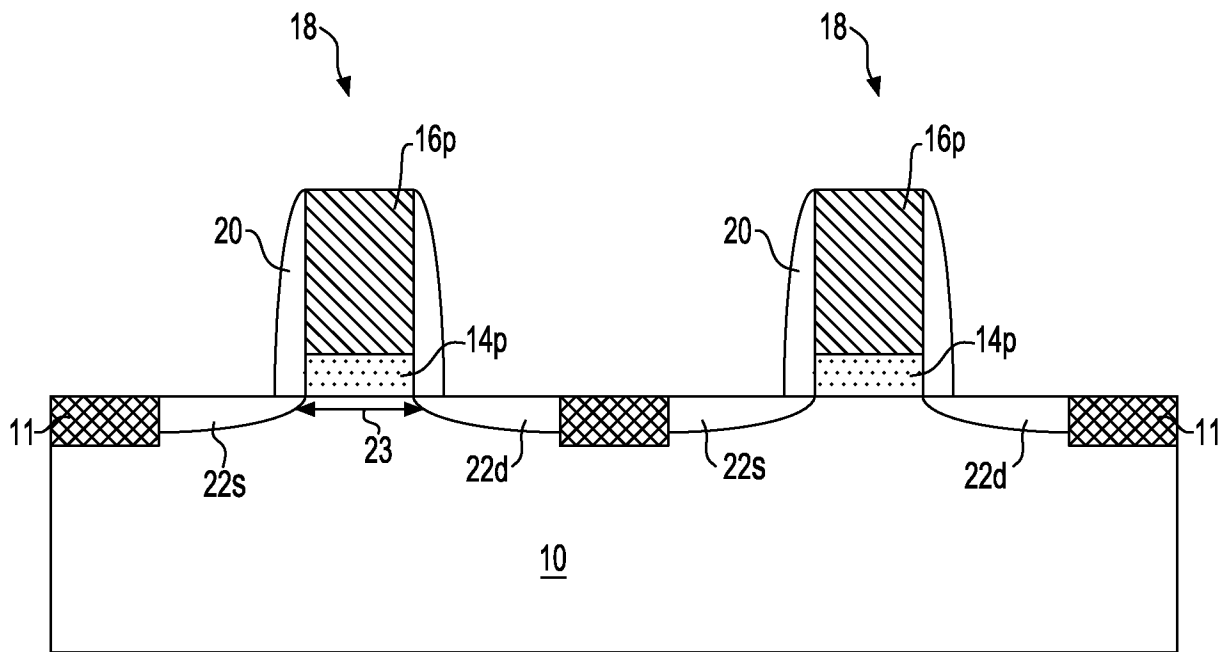


FIG. 3

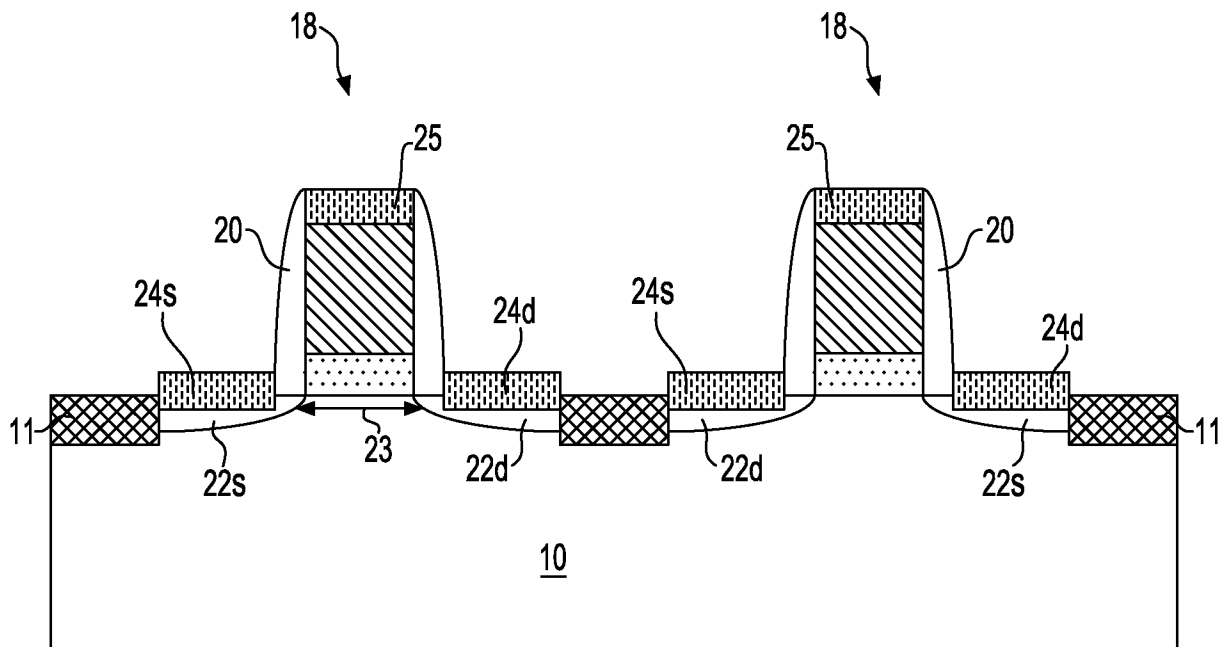


FIG. 4

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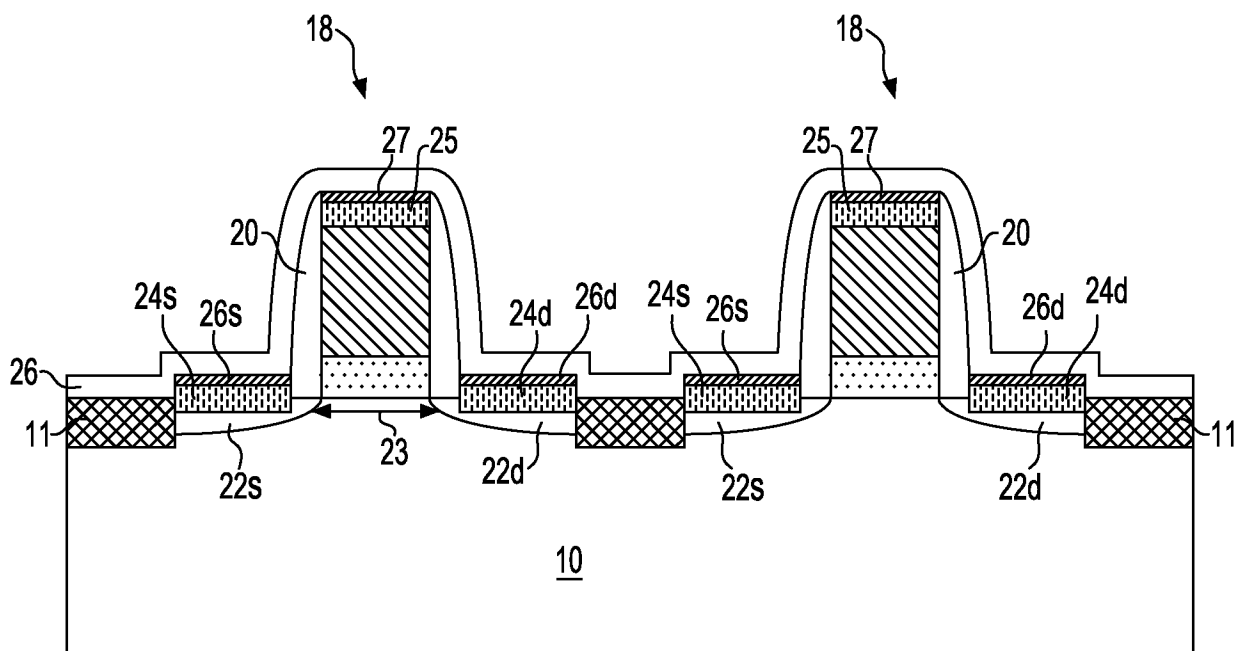


FIG. 5

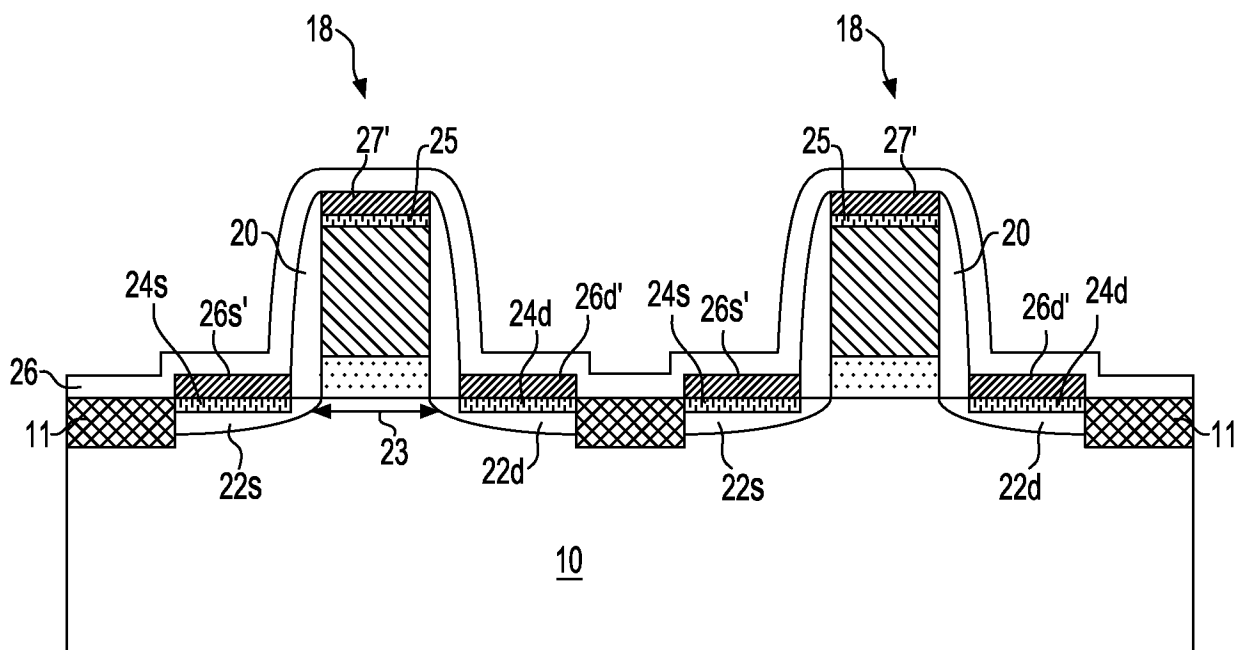


FIG. 6

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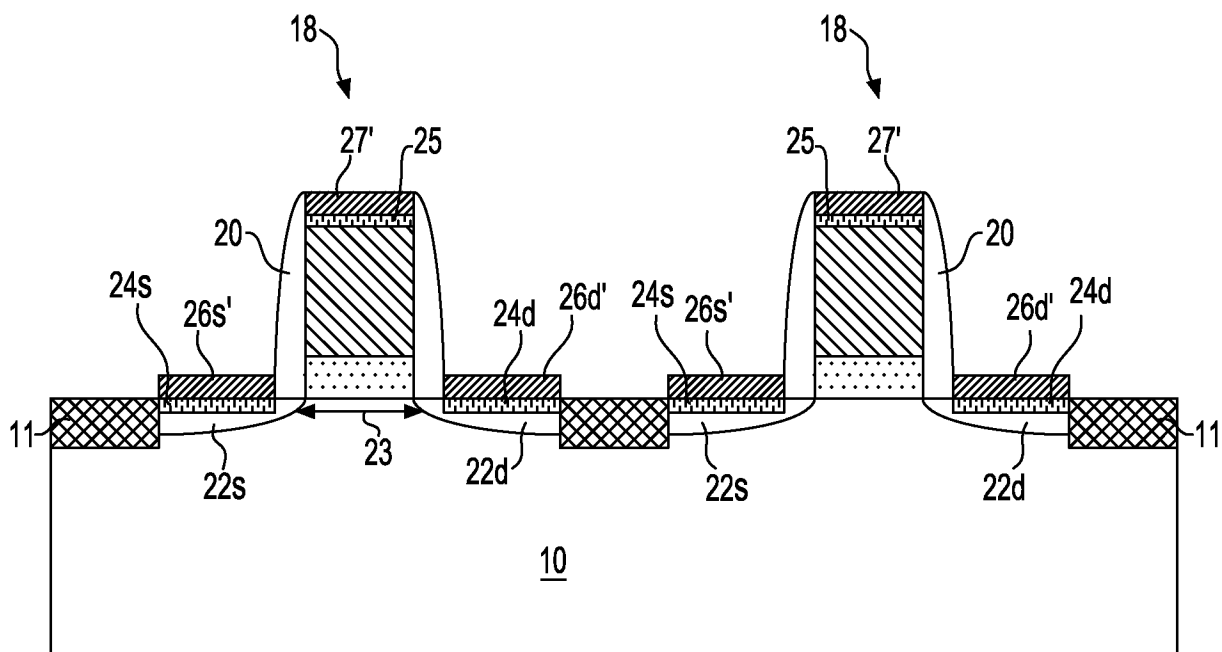


FIG. 7

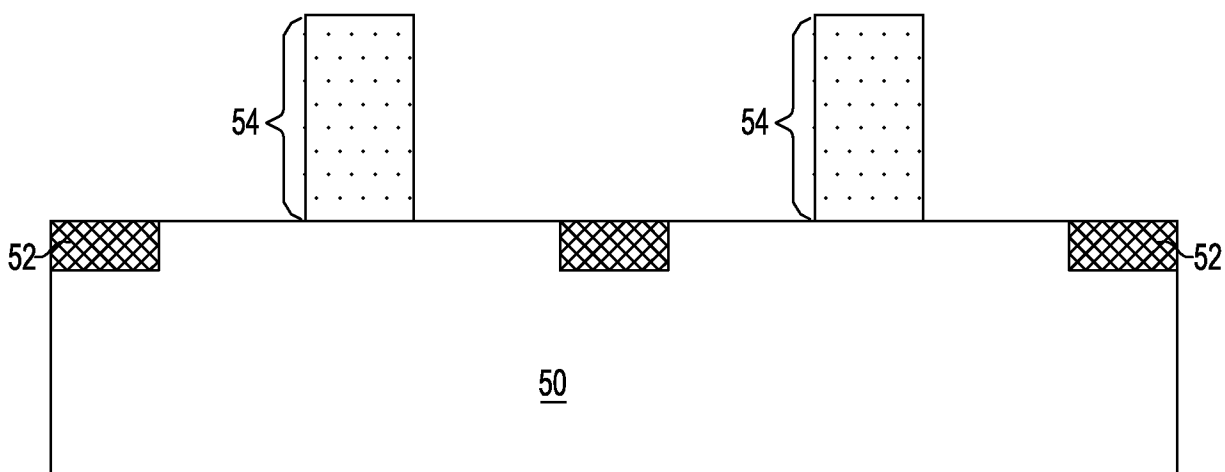


FIG. 8

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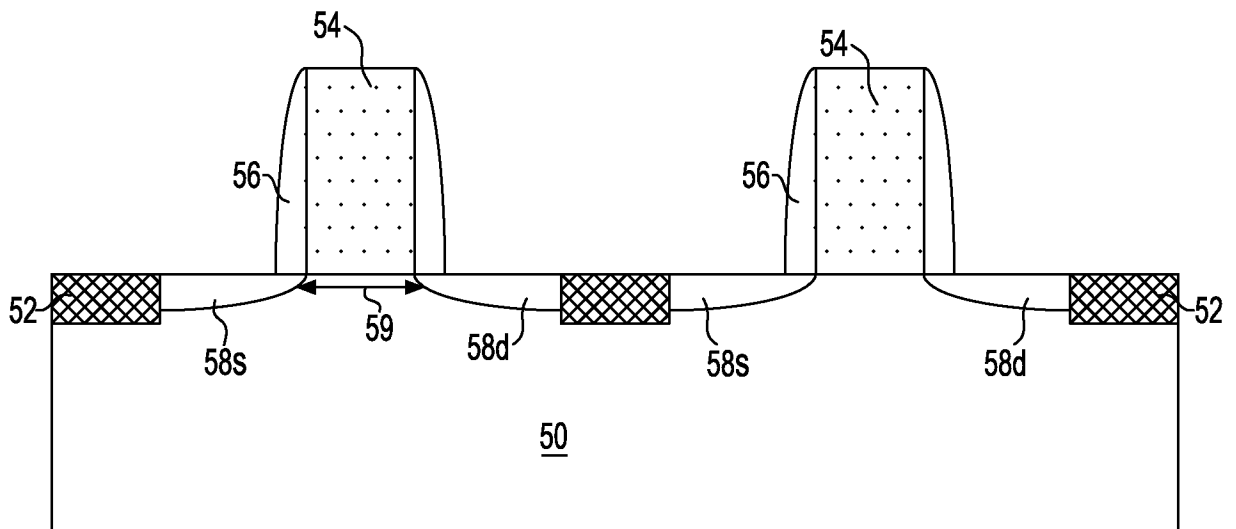


FIG. 9

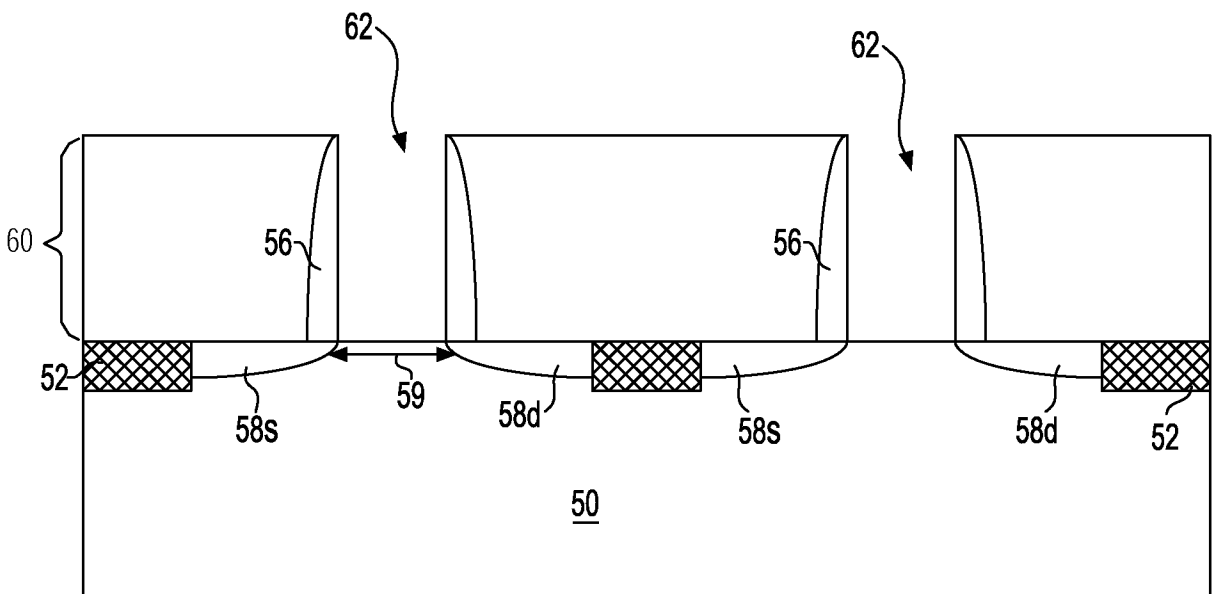


FIG. 10

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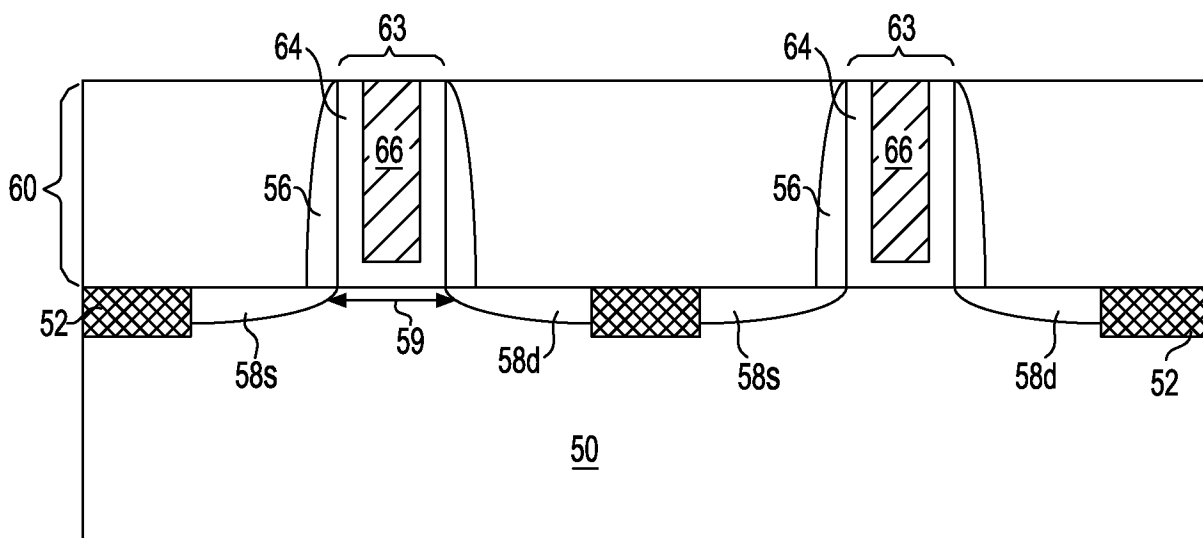


FIG. 11

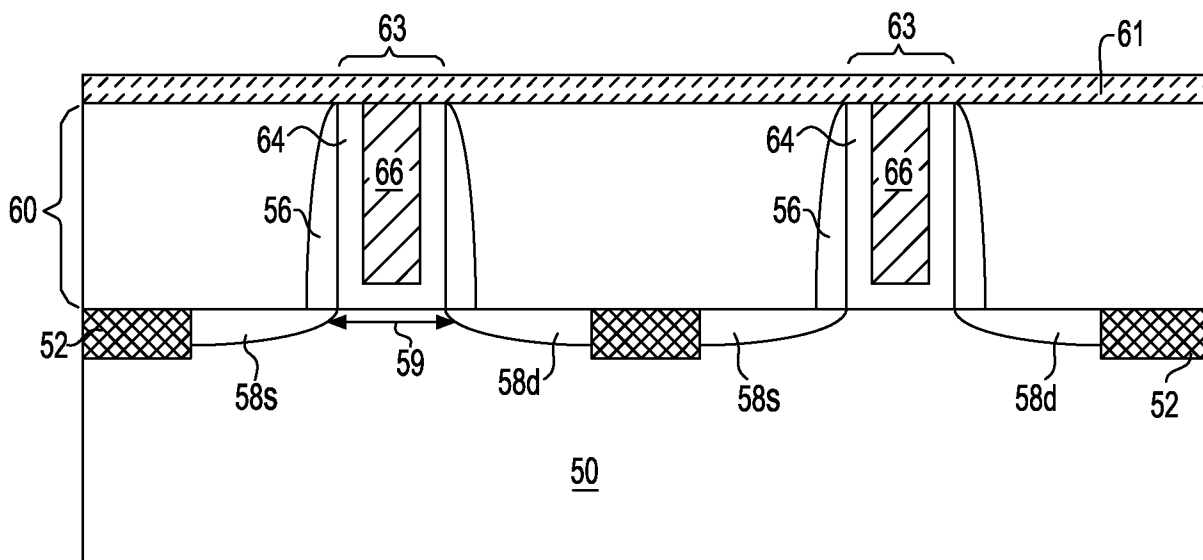


FIG. 12

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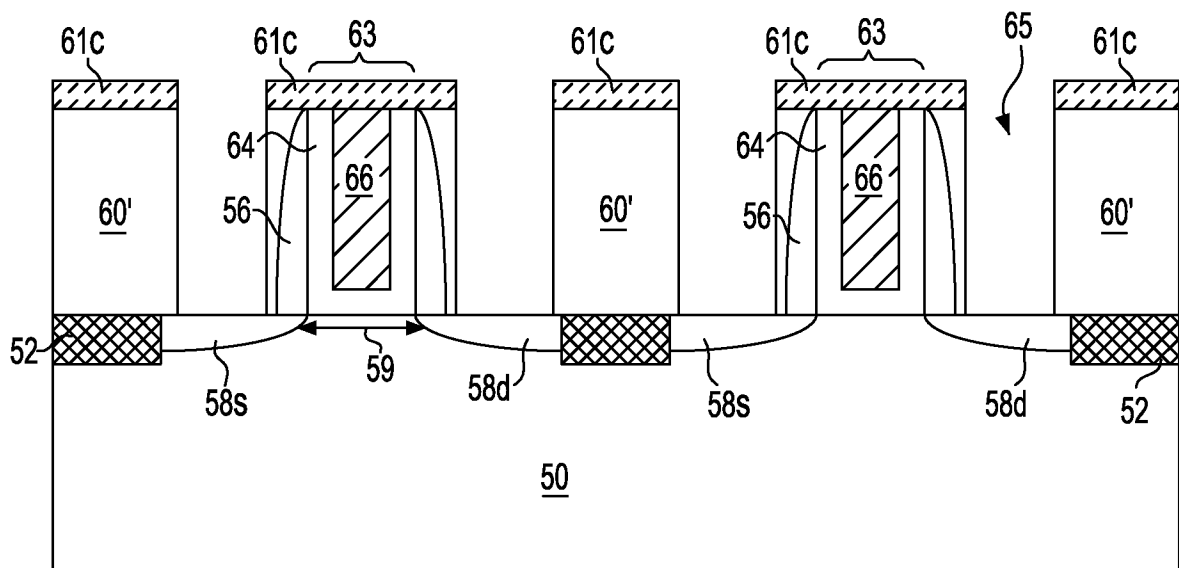


FIG. 13

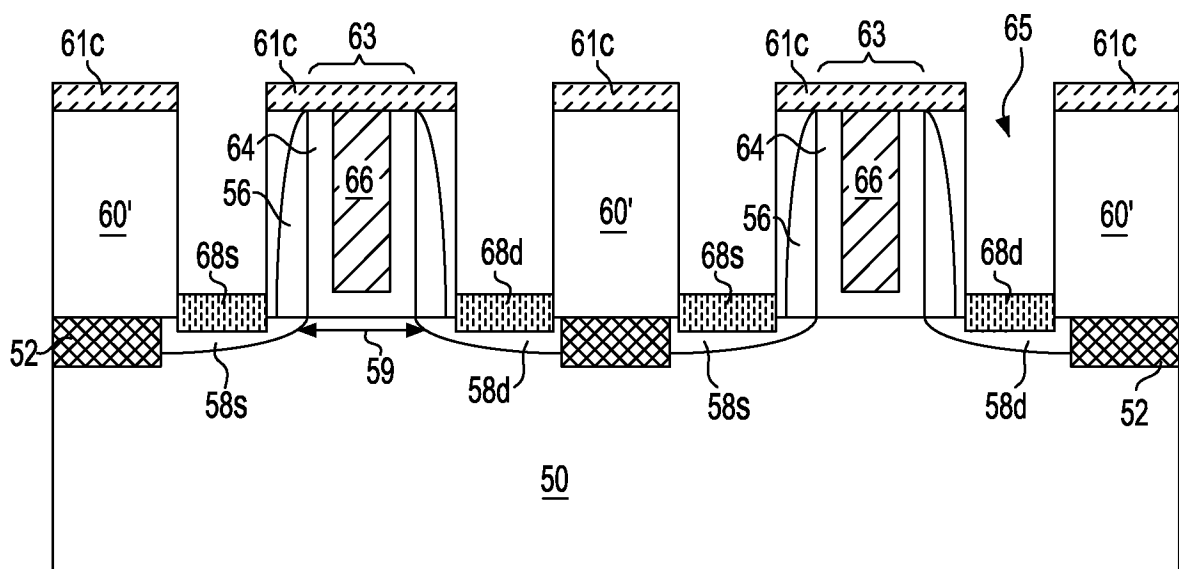


FIG. 14



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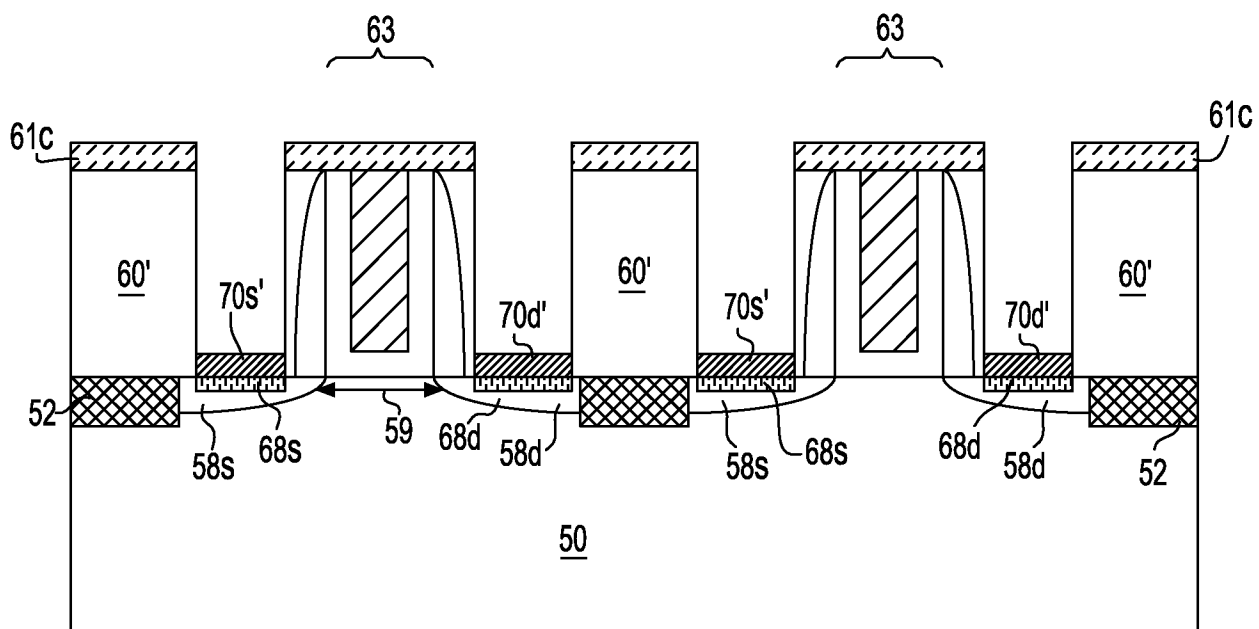


FIG. 17

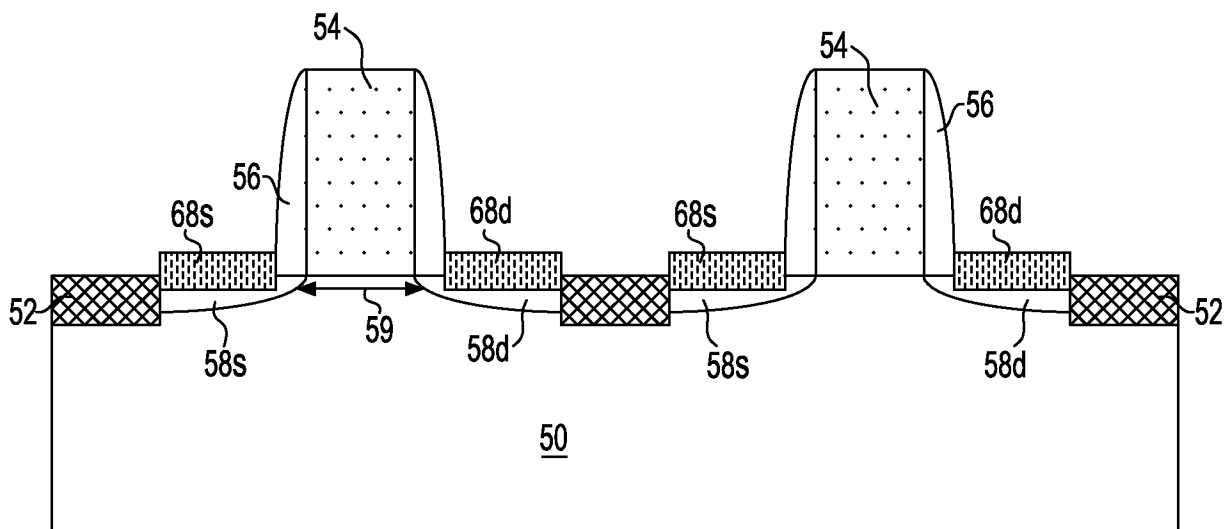


FIG. 18

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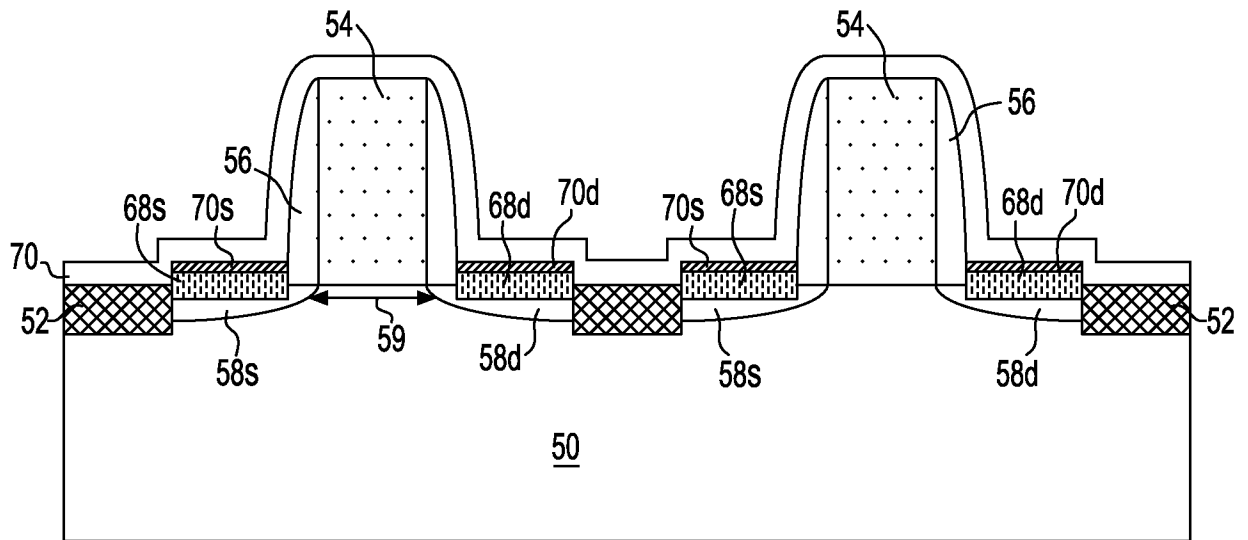


FIG. 19

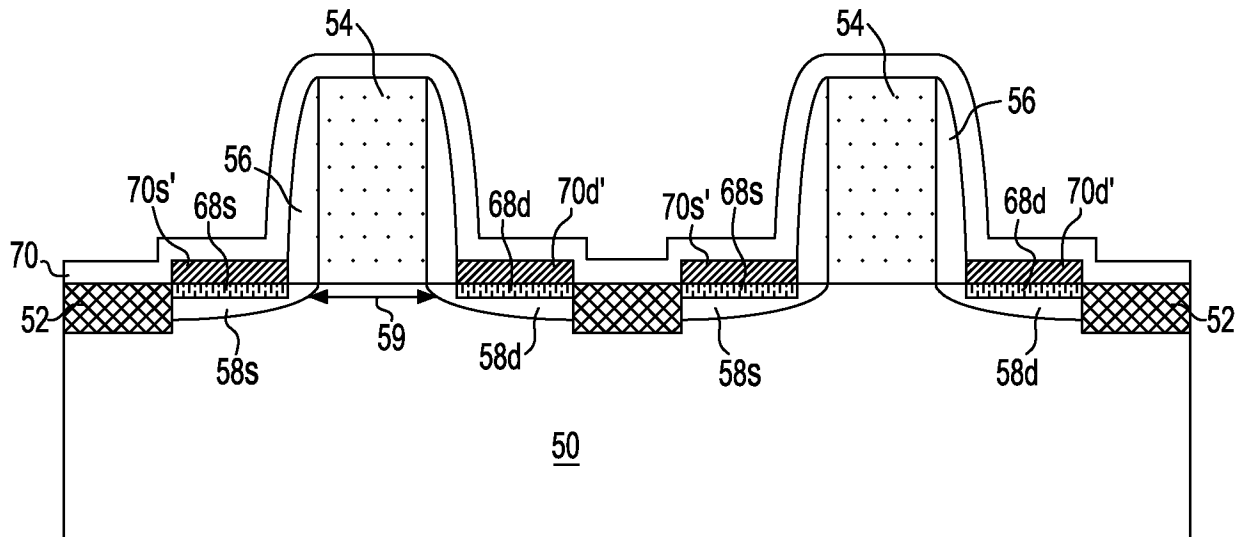


FIG. 20

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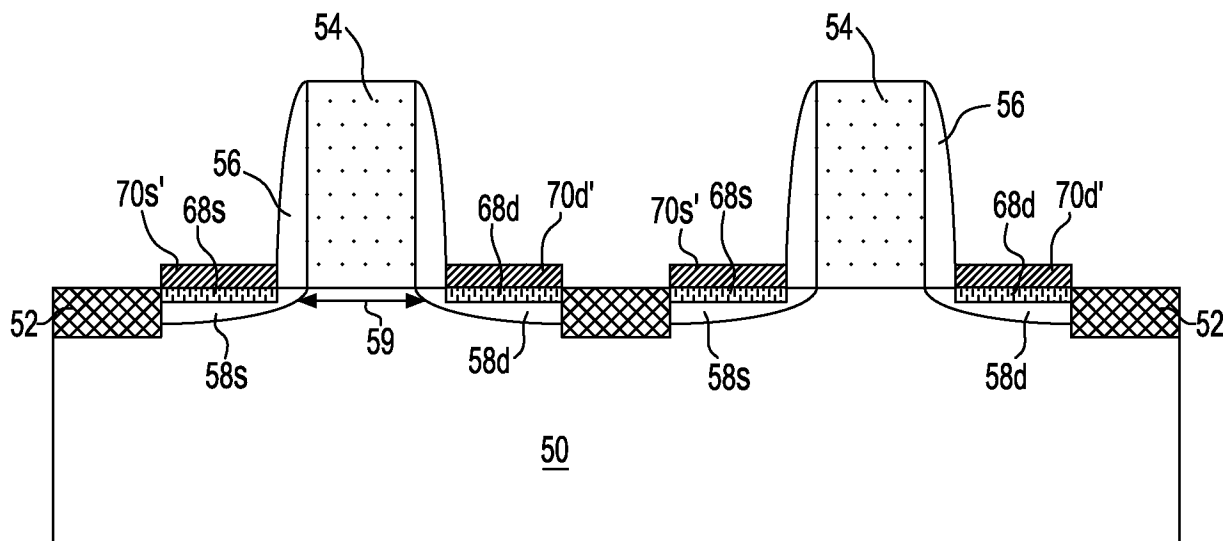


FIG. 21

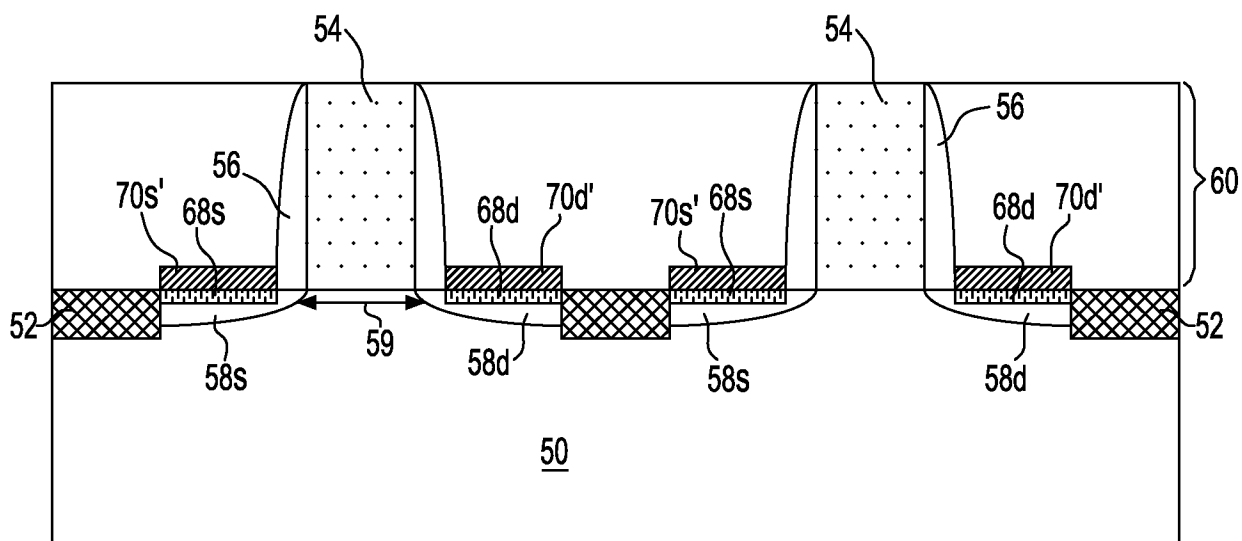


FIG. 22

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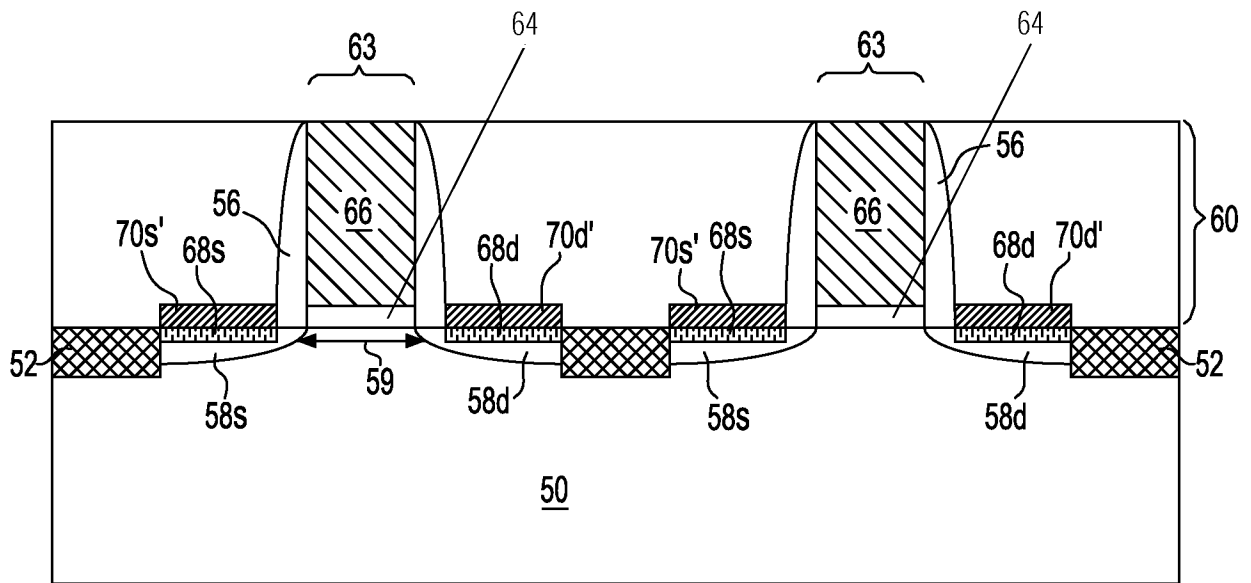


FIG. 23

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS

First Named Inventor/Applicant Name:

NICOLAS BREIL

Filer:

Margaret A. Pepper/hilda heinlein

Attorney Docket Number:

FIS920130015US1

Filed as Large Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	280	280
Utility Search Fee	1111	1	600	600
Utility Examination Fee	1311	1	720	720

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1600

Electronic Acknowledgement Receipt

EFS ID:	18171899
Application Number:	14177481
International Application Number:	
Confirmation Number:	5539
Title of Invention:	SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS
First Named Inventor/Applicant Name:	NICOLAS BREIL
Customer Number:	45600
Filer:	Margaret A. Pepper/hildah heinlein
Filer Authorized By:	Margaret A. Pepper
Attorney Docket Number:	FIS920130015US1
Receipt Date:	11-FEB-2014
Filing Date:	
Time Stamp:	13:41:05
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 1600
RAM confirmation Number	11936
Deposit Account	090458
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1	Application Data Sheet	FIS920130015US1_ADS.pdf	1112307	no	8
			6a21659a9b7222385b416a83197e78ba2a3a48a6		
Warnings:					
Information:					
2		FIS920130015US1_FiledApplica tion.pdf	160865	yes	40
			ff5e7e6d5e1419c4ac3526b2909dae4d785c9d80		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	34	
	Claims		35	39	
	Abstract		40	40	
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3	Power of Attorney	FIS920130015US1_powerofatt orney.pdf	211929	no	2
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4	Information Disclosure Statement (IDS) Form (SB08)	FIS920130015US1_SB08A.pdf	614955	no	5
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5	Drawings-only black and white line drawings	FIS920130015US1_formaldrawi ngs_EFS.pdf	509700	no	12
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6	Fee Worksheet (SB06)	fee-info.pdf	33056	no	2
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920130015US1(29690)
		Application Number	
Title of Invention	SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
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Inventor Information:

Inventor 1					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Nicolas		Breil		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Grenoble	Country of Residence i	FR		
Mailing Address of Inventor:					
Address 1	2070 Route 52				
Address 2					
City	Hopewell Junction	State/Province	NY		
Postal Code	12533	Country i	US		
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Prefix	Given Name	Middle Name	Family Name	Suffix	
	Christian		Lavoie		
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Mailing Address of Inventor:					
Address 1	1101 Kitchawan Road				
Address 2	Route 134 / P.O. Box 218				
City	Yorktown Heights	State/Province	NY		
Postal Code	10598	Country i	US		
Inventor 3					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Ahmet	S.	Ozcan		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920130015US1(29690)	
		Application Number		
Title of Invention	SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS			

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	FIS920130015US1(29690)
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Attorney Docket Number	FIS920130015US1(29690)	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	12	Suggested Figure for Publication (if any)	

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SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS

BACKGROUND

[0001] The present application relates to semiconductor structures and methods of forming the same. More particularly, the present application relates to semiconductor structures including a stack of, from bottom to top, a metal semiconductor alloy portion and a transition metal-metal semiconductor alloy portion located on each side of a functional gate structure. The present application also provides methods for forming such semiconductor structures.

[0002] Field effect transistors (FETs) are the basic building block of today's integrated circuits. Such transistors can be formed in conventional bulk substrates (such as silicon) or in semiconductor-on-insulator (SOI) substrates.

[0003] State of the art FETs can be fabricated by depositing a gate conductor over a gate dielectric and a semiconductor substrate. Generally, the FET fabrication process implements lithography and etching processes to define the gate structures. After providing the gate structures, source/drain extensions are formed into a portion of the semiconductor substrate and on both sides of each gate structure by ion implantation. Sometimes this implant is performed using a spacer to create a specific distance between the gate structure and the implanted junction. In some instances, such as in the manufacture of an n-FET device, the source/drain extensions for the n-FET device are implanted with no spacer. For a p-FET device, the source/drain extensions are typically implanted with a spacer present. A thicker spacer is typically formed after the source/drain extensions have been implanted. In some instances, deep source/drain implants can be performed with the thick spacer present. In other instances, and for advanced technologies, the source region and the drain region can be formed using a selective epitaxial growth process. High temperature anneals can be performed to activate the junctions after which the source/drain and top portion of the gate are generally converted into a metal semiconductor alloy (i.e., a metal silicide). The formation of the metal semiconductor alloy typically requires that a transition metal be deposited on the semiconductor substrate followed by a process to

produce the metal semiconductor alloy. Such a process forms low resistivity metal semiconductor alloy contacts to the deep source/drain regions.

[0004] In current technologies, and among other limiting factors, the metal semiconductor alloy containing contacts can be affected by the following two defects: (1) Encroachment of the metal semiconductor alloy which typically increases leakage through the source/drain junction and can also completely short the source region and the drain region of the device. (2) Spotty (non-uniform) metal semiconductor alloys which may cause unwanted etching of the metal semiconductor alloy contact during a subsequently performed etch.

[0005] These two defects are either formed during a subsequently performed etching process or the defects are enhanced by a subsequently performed etching process. Limiting these effects is critical for upcoming technologies.

SUMMARY

[0006] Metal semiconductor alloy contacts are provided on each of a source region and a drain region which are present in a semiconductor substrate. A transition metal is then deposited on each of the metal semiconductor alloy contacts, and during the deposition of the transition metal, the deposited transition metal reacts preferably, but not necessarily always, *in-situ* with a portion of each of the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact. Each transition metal-metal semiconductor alloy liner that is provided has outer edges that are vertically coincident, i.e., vertically aligned, with outer edges of each metal semiconductor alloy contact. The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy contact. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process.

[0007] In one aspect of the present application, semiconductor structures are provided. The semiconductor structures of the present application include a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of the semiconductor substrate, wherein the source region and the drain region are spaced apart by a channel region. The semiconductor structure also includes a functional gate structure located above the channel region, a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition metal-metal semiconductor alloy portion located on one side of the functional gate structure and located atop the source region, and a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of the functional gate structure and located atop the drain region.

[0008] In another aspect of the present application, methods of forming semiconductor structures are provided. In one embodiment, the method includes providing a source region and a drain region at a footprint of a functional gate structure. Next a source-side metal semiconductor alloy portion is formed on the source region, and a drain-side metal semiconductor alloy portion is formed on the drain region. A transition metal is then deposited on at least the source-side metal semiconductor alloy portion and the drain-side metal semiconductor alloy portion, wherein during the depositing the transition metal reacts preferably *in-situ* with the source-side metal semiconductor alloy portion forming a source-side transition metal-metal semiconductor alloy portion on a surface of the source-side metal semiconductor alloy portion, and the transition metal reacts preferably *in-situ* with the drain-side metal semiconductor alloy portion forming a drain-side transition metal-metal semiconductor alloy portion on a surface of the drain-side metal semiconductor alloy portion. Next, remaining portions of the transition metal are removed.

[0009] In another embodiment, the method includes providing a source region and a drain region at a footprint of a sacrificial gate structure. Next, a source-side metal semiconductor alloy portion is formed on the source region, and a drain-side metal semiconductor alloy portion is formed on the drain region. A transition metal is then deposited on at least the source-side metal semiconductor alloy portion and the drain-side metal semiconductor alloy portion, wherein

during the depositing the transition metal reacts preferably *in-situ* with the source-side metal semiconductor alloy portion forming a source-side transition metal-metal semiconductor alloy portion on a surface of the source-side metal semiconductor alloy portion, and the transition metal reacts preferably *in-situ* with the drain-side metal semiconductor alloy portion forming a drain-side transition metal-metal semiconductor alloy portion on a surface of the drain-side metal semiconductor alloy portion. Next, remaining portions of the transition metal are removed. In accordance with the present application, the sacrificial gate structure can be replaced with a functional gate structure prior to forming the source-side metal semiconductor alloy portion on the source region, and the drain-side metal semiconductor alloy portion on the drain region, or after removing the transition metal.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a cross sectional view of a first exemplary semiconductor structure including a material stack located on an upper surface of semiconductor substrate in accordance with an embodiment of the present application.

[0011] FIG. 2 is a cross sectional view of the first exemplary semiconductor structure of FIG. 1 after patterning the material stack into at least one functional gate structure.

[0012] FIG. 3 is a cross sectional view of the first exemplary semiconductor structure of FIG. 2 after forming a spacer on each vertical sidewall of the at least one functional gate structure and forming a source region in a semiconductor material portion of the semiconductor substrate on one side of the at least one functional gate structure and forming a drain region in another semiconductor material portion of the semiconductor substrate on another side of the at least one functional gate structure.

[0013] FIG. 4 is a cross sectional view of the first exemplary semiconductor structure of FIG. 3 after forming a source-side metal semiconductor alloy portion on the source region and a drain-side metal semiconductor alloy portion on the drain region.

[0014] FIG. 5 is a cross sectional view of the first exemplary semiconductor structure of FIG. 4 after deposition of a transition metal in which a source-side transition metal-metal semiconductor alloy portion forms preferably *in-situ* on the source-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion forms preferably *in-situ* on the drain-side metal semiconductor alloy portion.

[0015] FIG. 6 is a cross sectional view of the first exemplary semiconductor structure of FIG. 5 after annealing.

[0016] FIG. 7 is a cross sectional view of the first exemplary semiconductor structure of FIG. 6 after removing remaining portions of the transition metal.

[0017] FIG. 8 is a cross sectional view of a second exemplary semiconductor structure including at least one sacrificial gate structure formed on an upper surface of a semiconductor substrate in accordance with another embodiment of the present application.

[0018] FIG. 9 is a cross sectional view of the second exemplary semiconductor structure of FIG. 8 after forming a spacer on each vertical sidewall of the at least one sacrificial gate structure and forming a source region on one side of the at least one sacrificial gate structure and a drain region on another side of the at least one sacrificial gate structure.

[0019] FIG. 10 is a cross sectional view of the second exemplary semiconductor structure of FIG. 9 after forming a dielectric material and thereafter forming a gate cavity by removing the at least one sacrificial gate structure.

[0020] FIG. 11 is a cross sectional view of the second exemplary semiconductor structure of FIG. 10 after forming a gate dielectric portion and a gate conductor portion within the gate cavity.

[0021] FIG. 12 is a cross sectional view of the second exemplary semiconductor structure of FIG. 11 after forming a dielectric cap material on an upper surface of the dielectric material.

[0022] FIG. 13 is a cross sectional view of the second exemplary semiconductor structure of FIG. 12 after forming a plurality of openings through the dielectric cap material and the dielectric material exposing the source region and the drain region.

[0023] FIG. 14 is a cross sectional view of the second exemplary semiconductor structure of FIG. 13 after forming a source-side metal semiconductor alloy portion on the source region and a drain-side metal semiconductor alloy portion on the drain region.

[0024] FIG. 15 is a cross sectional view of the second exemplary semiconductor structure of FIG. 14 after deposition of a transition metal in which a source-side transition metal-metal semiconductor alloy portion forms preferably *in-situ* on the source-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion forms preferably *in-situ* on the drain-side metal semiconductor alloy portion.

[0025] FIG. 16 is a cross sectional view of the second exemplary semiconductor structure of FIG. 15 after annealing.

[0026] FIG. 17 is a cross sectional view of the second exemplary semiconductor structure of FIG. 16 after removing remaining portions of the transition metal.

[0027] FIG. 18 is a cross sectional view of the second exemplary semiconductor structure of FIG. 9 after forming a source-side metal semiconductor alloy portion on the source region and a drain-side metal semiconductor alloy portion on the drain region in accordance with an alternative embodiment of the present application.

[0028] FIG. 19 is a cross sectional view of the second exemplary semiconductor structure of FIG. 18 after deposition of a transition metal in which a source-side transition metal-metal

semiconductor alloy portion forms preferably *in-situ* on the source-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion forms preferably *in-situ* on the drain-side metal semiconductor alloy portion.

[0029] FIG. 20 is a cross sectional view of the second exemplary semiconductor structure of FIG. 19 after annealing.

[0030] FIG. 21 is a cross sectional view of the second exemplary semiconductor structure of FIG. 20 after removing remaining portions of the transition metal.

[0031] FIG. 22 is a cross sectional view of the second exemplary semiconductor structure of FIG. 21 after forming a dielectric material.

[0032] FIG. 23 is a cross sectional view of the second exemplary semiconductor structure of FIG. 22 after forming a gate cavity by removing the at least one sacrificial gate structure, and forming a gate dielectric portion and a gate conductor portion within the gate cavity.

DETAILED DESCRIPTION

[0033] The present application will now be described in greater detail by referring to the following discussion and drawings that accompany the present application. It is noted that the drawings of the present application are provided for illustrative purposes only and, as such, the drawings are not drawn to scale. It is also noted that like and corresponding elements are referred to by like reference numerals.

[0034] In the following description, numerous specific details are set forth, such as particular structures, components, materials, dimensions, processing steps and techniques, in order to provide an understanding of the various embodiments of the present application. However, it will be appreciated by one of ordinary skill in the art that the various embodiments of the present application may be practiced without these specific details. In other instances, well-known

structures or processing steps have not been described in detail in order to avoid obscuring the present application.

[0035] The present application can begin by first providing metal semiconductor alloy contacts on each of a source region and a drain region which are present in a semiconductor substrate. A transition metal can then be deposited atop each of the metal semiconductor alloy contacts. During the deposition of the transition metal, the deposited transition metal reacts preferably, but not necessarily always, *in-situ* with a portion of each of the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact. An optional anneal can be used to thicken the previously formed transition metal-metal semiconductor alloy liner. The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy contact. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process. Some specific examples of the formation of the aforementioned transition metal-metal semiconductor alloy liner atop a metal semiconductor alloy contact are now provided.

[0036] Referring first to FIG. 1, there is illustrated a first exemplary semiconductor structure including a material stack 12 located on an upper surface of semiconductor substrate 10 in accordance with an embodiment of the present application. The material stack 12 includes, from bottom to top, a gate dielectric material 14 and a gate conductor 16.

[0037] In some embodiments of the present application, the semiconductor substrate 10 can be a bulk semiconductor substrate. When a bulk semiconductor substrate is employed as semiconductor substrate 10, the bulk semiconductor substrate can be comprised of any semiconductor material including, but not limited to, Si, Ge, SiGe, SiC, SiGeC, and III/V compound semiconductors such as, for example, InAs, GaAs, and InP. Multilayers of these semiconductor materials can also be used as the semiconductor material of the bulk semiconductor. In one embodiment, the semiconductor substrate 10 can be comprised of a single crystalline semiconductor material, such as, for example, single crystalline silicon. In other

embodiments, the semiconductor substrate 10 may comprise a polycrystalline or amorphous semiconductor material.

[0038] In another embodiment, a semiconductor-on-insulator (SOI) substrate (not specifically shown) can be employed as the semiconductor substrate 10. Although not specifically shown, one skilled in the art understands that an SOI substrate includes a handle substrate, a buried insulator layer located on an upper surface of the handle substrate, and a semiconductor layer located on an upper surface of the buried insulator layer. The handle substrate provides mechanical support for the buried insulator layer and the semiconductor layer.

[0039] The handle substrate and the semiconductor layer of the SOI substrate may comprise the same, or different, semiconductor material. The term “semiconductor” as used herein in connection with the semiconductor material of the handle substrate and the semiconductor layer denotes any semiconductor material including, for example, Si, Ge, SiGe, SiC, SiGeC, and III/V compound semiconductors such as, for example, InAs, GaAs, or InP. Multilayers of these semiconductor materials can also be used as the semiconductor material of the handle substrate and the semiconductor layer. In one embodiment, the handle substrate and the semiconductor layer are both comprised of silicon. In some embodiments, the handle substrate is a non-semiconductor material including, for example, a dielectric material and/or a conductive material. In yet other embodiments, the handle substrate can be omitted and a substrate including an insulator layer and a semiconductor layer can be used as semiconductor substrate 10.

[0040] In some embodiments, the handle substrate and the semiconductor layer may have the same or different crystal orientation. For example, the crystal orientation of the handle substrate and/or the semiconductor layer may be {100}, {110}, or {111}. Other crystallographic orientations besides those specifically mentioned can also be used in the present application. The handle substrate and/or the semiconductor layer of the SOI substrate may be a single crystalline semiconductor material, a polycrystalline material, or an amorphous material. Typically, at least the semiconductor layer is a single crystalline semiconductor material. In

some embodiments, the semiconductor layer that is located atop the buried insulator layer can be processed to include semiconductor regions having different crystal orientations.

[0041] The buried insulator layer of the SOI substrate may be a crystalline or non-crystalline oxide or nitride. In one embodiment, the buried insulator layer is an oxide such as, for example, silicon dioxide. The buried insulator layer may be continuous or it may be discontinuous. When a discontinuous buried insulator region is present, the insulator region exists as an isolated island that is surrounded by semiconductor material.

[0042] The SOI substrate may be formed utilizing standard processes including for example, SIMOX (separation by ion implantation of oxygen) or layer transfer. When a layer transfer process is employed, an optional thinning step may follow the bonding of two semiconductor wafers together. The optional thinning step reduces the thickness of the semiconductor layer to a layer having a thickness that is more desirable.

[0043] In one example, the thickness of the semiconductor layer of the SOI substrate can be from 100 Å to 1000 Å. In another example, the thickness of the semiconductor layer of the SOI substrate can be from 500 Å to 700 Å. In some embodiments, and when an ETSOI (extremely thin semiconductor-on-insulator) substrate is employed, the semiconductor layer of the SOI has a thickness of less than 100 Å. If the thickness of the semiconductor layer is not within one of the above mentioned ranges, a thinning step such as, for example, planarization or etching can be used to reduce the thickness of the semiconductor layer to a value within one of the ranges mentioned above. The buried insulator layer of the SOI substrate typically has a thickness from 10 Å to 2000 Å, with a thickness from 1000 Å to 1500 Å being more typical. The thickness of the handle substrate of the SOI substrate is inconsequential to the present application.

[0044] In some other embodiments, hybrid semiconductor substrates which have different surface regions of different crystallographic orientations can be employed as semiconductor substrate 10. When a hybrid substrate is employed, an nFET is typically formed on a (100) crystal surface, while a pFET is typically formed on a (110) crystal plane. The hybrid substrate

can be formed by techniques that are well known in the art. See, for example, U.S. Patent No. 7,329,923, U.S. Publication No. 2005/0116290, dated June 2, 2005 and U.S. Patent No. 7,023,055, the entire contents of each are incorporated herein by reference.

[0045] Semiconductor substrate 10 may be doped, undoped or contain doped and undoped regions therein. For clarity, the doped regions are not specifically shown in the drawings of the present application. Each doped region within the semiconductor material of the semiconductor substrate 10 may have the same, or they may have different conductivities and/or doping concentrations. The doped regions that are present in the semiconductor material of semiconductor substrate 10 are typically referred to as well regions and they are formed utilizing a conventional ion implantation process, gas phase doping or epitaxial growth.

[0046] In some embodiments, the semiconductor substrate 10 can be processed to include at least one isolation region 11 therein. The at least one isolation region 11 can be a trench isolation region or a field oxide isolation region. The trench isolation region can be formed utilizing a conventional trench isolation process well known to those skilled in the art. For example, lithography, etching and filling of the trench with a trench dielectric such as an oxide may be used in forming the trench isolation region. Optionally, a liner may be formed in the trench prior to trench fill, a densification step may be performed after the trench fill and a planarization process may follow the trench fill as well. The field oxide isolation region may be formed utilizing a so-called local oxidation of silicon process. Note that the at least one isolation region 11 may provide isolation between neighboring gate structures, typically required when the neighboring gate structures have opposite conductivities, i.e., n-type transistors and p-type transistors. As such, the at least one isolation region 11 can separate a first device region in which an n-type transistor device can be formed and second device region in which a p-type transistor device can be formed.

[0047] A material stack 12 is formed on an upper surface of the semiconductor substrate 10 and on an upper surface of each isolation region 11. As stated above, the material stack 12 includes, from bottom to top, a gate dielectric material 14 and a gate conductor 16. The gate dielectric

material 14 and the gate conductor 16 which provide material stack 12 are formed as blanket layers atop the upper surface of the semiconductor substrate 10.

[0048] In this embodiment of the present application, the material stack 12 is used in providing a functional gate structure. The term “functional gate structure” is used throughout the present application as a permanent gate structure used to control output current (i.e., flow of carriers in the channel) of a semiconducting device through electrical or magnetic fields.

[0049] The gate dielectric material 14 that can be used in the present application can be a semiconductor oxide, a semiconductor nitride, and/or a semiconductor oxynitride. In one example, the gate dielectric material 14 can be composed of silicon dioxide, silicon nitride and/or silicon oxynitride. In another embodiment of the present application, the gate dielectric material 14 may include at least a dielectric metal oxide. Exemplary dielectric metal oxides that can be used as gate dielectric material 14 include, but are not limited to, HfO_2 , ZrO_2 , La_2O_3 , Al_2O_3 , TiO_2 , SrTiO_3 , LaAlO_3 , Y_2O_3 , HfO_xN_y , ZrO_xN_y , $\text{La}_2\text{O}_x\text{N}_y$, $\text{Al}_2\text{O}_x\text{N}_y$, TiO_xN_y , SrTiO_xN_y , LaAlO_xN_y , $\text{Y}_2\text{O}_x\text{N}_y$, SiON , SiN_x , a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3 and each value of y is independently from 0 to 2. In some embodiments, a multilayered gate dielectric structure comprising different gate dielectric materials, e.g., silicon dioxide, and a dielectric metal oxide can be formed and used as the gate dielectric material 14.

[0050] In some embodiments of the present application, the gate dielectric material 14 can be formed by a deposition technique such as, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), physical vapor deposition (PVD), sputtering, or atomic layer deposition. In another embodiment of the present application, the gate dielectric material 14 can be formed by a thermal growth technique such as, for example, thermal oxidation and/or thermal nitridation. In yet a further embodiment of the present application, a combination of a deposition and thermal growth may be used in forming a multilayered gate dielectric structure.

[0051] In one embodiment of the present application, the gate dielectric material 14 can have a thickness in a range from 1 nm to 10 nm. Other thicknesses that are lesser than or greater than the aforementioned thickness range can also be employed for the gate dielectric material 14.

[0052] The gate conductor 16 that can be used in the present application can be composed of doped polysilicon, doped silicon germanium, an elemental metal (e.g., tungsten, titanium, tantalum, aluminum, nickel, ruthenium, palladium and platinum), an alloy of at least two elemental metals, an elemental metal nitride (e.g., tungsten nitride, aluminum nitride, and titanium nitride), or multilayered combinations thereof. In one embodiment, an entirety of the gate conductor 16 is comprised of a doped polysilicon or doped polysilicon germanium. In another embodiment, a lower portion of the gate conductor 16 is comprised a conductive material other than doped polysilicon or doped polysilicon germanium, and an upper portion of the gate conductor 16 is comprised of doped polysilicon or doped silicon germanium.

[0053] The gate conductor 16 can be formed utilizing a deposition process including, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), physical vapor deposition (PVD), sputtering, atomic layer deposition (ALD) or other like deposition processes. In one embodiment, the gate conductor 16 has a thickness from 1 nm to 100 nm. Other thicknesses that are lesser than or greater than the aforementioned thickness range can also be employed for the gate conductor 16.

[0054] Referring now to FIG. 2, there is illustrated the first exemplary semiconductor structure of FIG. 1 after patterning the material stack 12 (including gate dielectric material 14 and gate conductor 16) into at least one functional gate structure 18. The at least one functional gate structure 18 includes, from bottom to top, a gate dielectric portion 14p and a gate electrode portion 16p. Gate dielectric portion 14p includes a remaining portion of the gate dielectric material 14 of material stack 12, while gate conductor portion 16p includes a remaining portion of the gate conductor 16 of material stack 12.

[0055] The at least one functional gate structure 18 can be formed by lithography and etching. Lithography can include forming a photoresist (not shown) on the topmost surface of gate conductor 16 exposing the photoresist to a desired pattern of radiation, and then developing the exposed photoresist with a resist developer to provide a patterned photoresist atop the gate conductor 16 of material stack 12. At least one etch is then employed which transfers the pattern from the patterned photoresist into the various materials. In one embodiment, the etch used for pattern transfer may include a dry etch process such as, for example, reactive ion etching, plasma etching, ion beam etching or laser ablation. In another embodiment, the etch used for pattern transfer may include a wet chemical etchant such as, for example, KOH (potassium hydroxide). In yet another embodiment, a combination of a dry etch and a wet chemical etch may be used to transfer the pattern. After transferring the pattern into the material layers, the patterned photoresist can be removed utilizing a resist stripping process such as, for example, ashing.

[0056] As is shown in the embodiment illustrated in FIG. 2, the sidewall edges of the gate dielectric material portion 14p and the gate conductor 16p are vertically coincident to (i.e., vertically aligned with) each other.

[0057] Referring now to FIG. 3, there is illustrated the first exemplary semiconductor structure of FIG. 2 after forming a spacer 20 on each vertical sidewall of the at least one functional gate structure 18 and forming a source region 22s in a semiconductor material portion of the semiconductor substrate 10 on one side of the at least one functional gate structure 18 and forming a drain region 22d in another semiconductor material portion of the semiconductor substrate 10 on another side of the at least one functional gate structure 18. The semiconductor portion of the semiconductor substrate 10 that is located beneath the functional gate structure 18 and positioned between the source region 22s and the drain region 22d is referred as a channel region 23.

[0058] The spacer 20 can be formed by first providing a spacer material and then etching the spacer material. The spacer material may be composed of any dielectric spacer material including, for example, a dielectric oxide, dielectric nitride, and/or dielectric oxynitride. In one

example, the spacer material used in providing the spacer 20 may be composed of silicon dioxide or silicon nitride. The spacer material can be provided by a deposition process including, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), or physical vapor deposition (PVD). The etching of the spacer material may comprise a dry etch process such as, for example, a reactive ion etch.

[0059] FIG. 3 also shows the presence of a source region 22s located within a semiconductor material portion of the semiconductor substrate 10 on one side of the at least one functional gate structure 18 and a drain region 22d located within another semiconductor material portion of the semiconductor substrate 10 on another side of the at least one functional gate structure 18. In some embodiments, a source extension region and drain extension region (not separately shown) are typically formed prior to forming the spacer 20 utilizing an extension ion implantation process. As such, a portion of each of the source extension region and drain extension region would be located beneath the spacer 20. After forming the spacer 20, source region 22s and drain region 22d are formed utilizing a source/drain ion implantation process. An activation anneal may follow the implantation processes.

[0060] The source region 22s (including the corresponding source extension region) and the drain region (including the corresponding drain extension region) may be doped with a p-type or n-type dopant. The term "p-type" refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. In a silicon-containing substrate, examples of p-type dopants, i.e., impurities, include, but are not limited to, boron, aluminum, gallium and indium. "N-type" refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. In a silicon containing substrate examples of n-type dopants, i.e., impurities, include, but are not limited to, antimony, arsenic and phosphorous. The concentration of dopants within the source region 22s (and corresponding source extension region) and the drain region 22d (and the corresponding drain extension region) can be within ranges typically used in forming metal oxide semiconductor field effect transistors (MOSFETs).

[0061] Referring now to FIG. 4, there is illustrated the first exemplary semiconductor structure of FIG. 3 after forming a source-side metal semiconductor alloy portion 24s on the source region 22s and a drain-side metal semiconductor alloy portion 24d on the drain region 22d. The source-side metal semiconductor alloy portion 24s, and the drain-side metal semiconductor alloy portion 24d are comprised of a same metal semiconductor alloy.

[0062] In some embodiments such as shown in FIG. 4 and when a semiconductor material is present at an upper surface of the gate conductor portion 18p, a metal semiconductor alloy cap portion 25 can be formed on the gate conductor portion 18p. In some embodiments, the source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d and optionally the metal semiconductor alloy cap portion 25 each include a same metal semiconductor alloy. In another embodiment, the source-side metal semiconductor alloy portion 24s, and the drain-side metal semiconductor alloy portion 24d each comprise a first metal semiconductor alloy, while the optional metal semiconductor alloy cap portion 25 comprises a second metal semiconductor alloy that is different from the first metal semiconductor alloy. The difference between the first metal semiconductor alloy and the second metal semiconductor alloy is in respect to the semiconductor element, not the metallic element that forms each metal semiconductor alloy.

[0063] The source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d and optionally the metal semiconductor alloy cap portion 25, are formed by first providing a layer of a metal semiconductor alloy forming metal (not shown) on all exposed surfaces of the structure shown in FIG. 3. The term “metal semiconductor alloy forming metal” is used throughout the present application to denote a metal that can react with an underlying semiconductor material to form a metal semiconductor alloy. Illustrative examples of metal semiconductor alloy forming metals that can be used in the present application include at least one of nickel (Ni), platinum (Pt), palladium (Pd), titanium (Ti), tungsten (W), and cobalt (Co). The metal semiconductor alloy forming metal can be formed by a deposition process including, for example, chemical vapor deposition, plasma enhanced chemical vapor deposition, plating or sputtering. In some embodiments, a co-deposition of metal semiconductor alloy

forming metals can be used. In another embodiment, a first metal semiconductor alloy forming metal can be formed, followed by a second metal semiconductor alloy forming metal. The metal semiconductor alloy metal that is formed can have a thickness from 5 to 15 nm. Other thicknesses that are greater than or lesser than the aforementioned thickness range can also be employed as the thickness of the metal semiconductor alloy forming metal.

[0064] After providing the metal semiconductor alloy forming metal and in some embodiments of the present application, a diffusion barrier (not shown) can be formed on an exposed upper surface of the metal semiconductor alloy forming metal. In another embodiment of the present application, no diffusion barrier is provided on the exposed upper surface of the metal semiconductor alloy forming metal. When present, the diffusion barrier can include a metal nitride such as, for example, TiN or TaN, and any deposition process including those mentioned above for providing the metal semiconductor alloy forming metal may be used. When present, the diffusion barrier can have a thickness from 1 nm to 20 nm.

[0065] Next, an anneal is performed under conditions that are effective in causing the metal semiconductor alloy forming metal to diffuse into the semiconductor material portions of the source region 22s, the drain region 22d and optionally the gate conductor portion 18p, forming a metal semiconductor alloy. Notably, the anneal can form the source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d and optionally the metal semiconductor alloy cap portion 25. In one embodiment, the source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d and optionally the metal semiconductor alloy cap portion 25 each comprise a metal silicide such as, for example, nickel silicide or platinum-nickel silicide. In another embodiment, the source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d and optionally the metal semiconductor alloy cap portion 25 each include a metal germanide such as nickel germanide.

[0066] The metal semiconductor alloy formation anneal may be performed in a single step or a two-step anneal can be used. In one embodiment and when nickel is used, the metal semiconductor alloy formation anneal can be performed at a temperature of from 200 °C to 500

°C. In another embodiment, temperatures greater than 500°C can be used. The metal semiconductor alloy formation anneal is typically performed in an ambient including, for example, argon, helium, neon and/or nitrogen. The metal semiconductor alloy formation anneal can be performed utilizing a rapid thermal anneal, a spike anneal, a microwave anneal or a laser anneal. Following the metal semiconductor alloy formation anneal, the optional diffusion barrier and any unreacted metal semiconductor alloy forming metal can be removed utilizing one or more etch processes.

[0067] The above processing forms a source-side metal semiconductor alloy portion 24s that has outer edges that are vertically aligned with outer edges of the source region 22s, a drain-side metal semiconductor alloy portion 24d that has outer edges that are vertical aligned with outer edges of the drain region 22d, and an optional metal semiconductor alloy cap portion 25 that has outer edges that are vertically aligned with outer edges of a remaining portion of the gate conductor portion 16p. In some embodiments of the present application, the metal semiconductor alloy formation anneal may partially or entirely consume the semiconductor material portion of the source region 22s and the drain region 22d.

[0068] Referring now to FIG. 5, there is illustrated the first exemplary semiconductor structure of FIG. 4 after deposition of a transition metal 26 in which a source-side transition metal-metal semiconductor alloy portion 26s forms preferably *in-situ* on the source-side metal semiconductor alloy portion 24s and a drain-side transition metal-metal semiconductor alloy portion 26d forms preferably *in-situ* on the drain-side metal semiconductor alloy portion 24d. An optional transition metal-metal semiconductor alloy cap portion 27 can also be formed preferably *in-situ* on the metal semiconductor alloy cap portion 25. The source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 are formed simultaneously with the deposition of transition metal 26 by diffusion and intermixing of the transition metal with the underlying source-side metal semiconductor alloy portion 24s, drain-side metal semiconductor alloy portion 24d, and metal semiconductor alloy cap portion 25, respectively.

[0069] The transition metal 26 that provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 comprises any transition metal that can diffuse and interact with the underlying source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d and, optionally, the metal-metal semiconductor alloy cap portion 25, with the proviso that the transition metal 26 is different from the metal semiconductor alloy forming metal mentioned above. Examples of transition metals that may be employed at this point of the present application include transition metals from Groups 4, 5 and 6 of the Periodic Table such as, for example, titanium (Ti), zirconium (Zr), hafnium (Hf), tantalum (Ta), tungsten (W), niobium (Nb), molybdenum (Mo), vanadium (V), and chromium (Cr). Rhenium (Re) can also be used as the transition metal 26. In some embodiments in which nFETs and pFETs are formed, different compositions of transition metals can be used in forming the source-side transition metal-metal semiconductor alloy portion 26s and the drain-side transition metal-metal semiconductor alloy portion 26d.

[0070] The transition metal 26 that provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 can be formed by any of the deposition processes mentioned above in forming the metal semiconductor alloy forming metal. In one example, the transition metal 26 that provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 can be formed by sputtering. In one embodiment, the sputtering can be performed at a temperature from room temperature (i.e., 20°C) up to, and including 500°C. The thickness of the transition metal 26 that provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 can be from 2 nm to 10 nm. Other thicknesses can also be employed for the thickness of the transition metal 26 that

provides the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27.

[0071] In accordance with the present application, the source-side transition metal-metal semiconductor alloy portion 26s is composed on an intermixed product of the transition metal 26 and the previously formed source-side metal semiconductor alloy portion 24s. Typically, the source-side transition metal-metal semiconductor alloy portion 26s can contain up to 70% of the transition metal 26. In one example, and when the transition metal 26 comprises titanium, and the source-side metal semiconductor alloy portion 24s comprises nickel silicide, then the source-side transition metal-metal semiconductor alloy portion 26s comprises titanium-nickel silicide. In some embodiments, the source-side transition metal-metal semiconductor alloy portion 26s may comprise titanium-platinum-nickel silicide. The source-side transition metal-metal semiconductor alloy portion 26s has outer edges that are vertically aligned with outer edges of the source-side metal semiconductor alloy portion 24s. In one embodiment, the source-side transition metal-metal semiconductor alloy portion 26s has a first thickness from 1 nm to 10 nm. In another embodiment, the source-side transition metal-metal semiconductor alloy portion 26s has a first thickness from 1 nm to 3 nm.

[0072] In accordance with the present application, the drain-side transition metal-metal semiconductor alloy portion 26d is composed on an intermixed product of the transition metal 26 and the previously formed drain-side metal semiconductor alloy portion 24d. Typically, the drain-side transition metal-metal semiconductor alloy portion 26d can contain up to 70% of the transition metal 26. In one example, and when the transition metal 26 comprises titanium, and the drain-side metal semiconductor alloy portion 24d comprises nickel silicide, then the drain-side transition metal-metal semiconductor alloy portion 26d comprises titanium-nickel silicide. In some embodiments, the drain-side transition metal-metal semiconductor alloy portion 26d may comprise titanium-platinum-nickel silicide. The drain-side transition metal-metal semiconductor alloy portion 26d has outer edges that are vertically aligned with outer edges of the drain-side metal semiconductor alloy portion 24d. In one embodiment, the drain-side

transition metal-metal semiconductor alloy portion 26d has a first thickness from 1 nm to 10 nm. In one embodiment, the drain-side transition metal-metal semiconductor alloy portion 26d has a first thickness from 1 nm to 3 nm. The source-side metal semiconductor-alloy portion 24s and the drain-side metal semiconductor alloy portion 24d that remain after forming source-side transition metal-metal semiconductor alloy portion 26s and the drain-side transition metal-metal semiconductor alloy portion 26d can have a thickness from 3 nm to 25 nm, with a thickness from 10 nm to 20 nm being more typical.

[0073] In accordance with the present application, the transition metal-metal semiconductor alloy cap portion 27 is composed on an intermixed product of the transition metal 26 and the previously formed metal semiconductor alloy cap portion 25. In one example, and when the transition metal 26 comprises titanium, and the metal semiconductor alloy cap portion 25 comprises nickel silicide, then the transition metal-metal semiconductor alloy cap portion 27 comprises titanium-nickel silicide. In some embodiments, transition metal-metal semiconductor alloy cap portion 27 may comprise titanium-platinum-nickel silicide. The transition metal-metal semiconductor alloy cap portion 27 has outer edges that are vertically aligned with outer edges of the metal semiconductor alloy cap portion 25. In one embodiment, the transition metal-metal semiconductor alloy cap portion 27 has a first thickness from 1 nm to 10 nm. In one embodiment, the transition metal-metal semiconductor alloy cap portion 27 has a first thickness from 1 nm to 3 nm.

[0074] The source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27 have a higher etch resistance (in either a wet etch or a dry etch process) than the source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d, and optionally the metal semiconductor alloy cap portion 25.

[0075] It is noted that the remaining portions of the source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d, and optionally the metal semiconductor alloy cap portion 25 have a thickness that is less than the thickness of each of the

source-side metal semiconductor alloy portion 24s, the drain-side metal semiconductor alloy portion 24d, and optionally the metal semiconductor alloy cap portion 25 prior to the formation of the source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27.

[0076] Referring now to FIG. 6, there is illustrated the first exemplary semiconductor structure of FIG. 5 after annealing. This anneal step is optional and need not be performed in all instances. The anneal process serves to thicken the previously formed source-side transition metal-metal semiconductor alloy portion 26s, the drain-side transition metal-metal semiconductor alloy portion 26d, and optionally the transition metal-metal semiconductor alloy cap portion 27. The thickened source-side transition metal-metal semiconductor alloy portion is labeled as element 26s', the thickened drain-side transition metal-metal semiconductor alloy portion is labeled as 26d', and the optional thickened transition metal-metal semiconductor alloy cap portion is labeled as 27' in FIG. 6.

[0077] Notably, the anneal provides a thickened source-side transition metal-metal semiconductor alloy portion 26s' having a second thickness that is greater than the first thickness of the previously formed source-side transition metal-metal semiconductor alloy portion 26s, a thickened drain-side transition metal-metal semiconductor alloy portion 26d' having a second thickness that is greater than the first thickness of the previously formed drain-side transition metal-metal semiconductor alloy portion 26d, and, if present, the thickened transition metal-metal semiconductor alloy cap portion 27' has a second thickness that is greater than the previously formed transition metal-metal semiconductor alloy cap portion 27. During this annealing, the source-side metal semiconductor alloy portion 24s and the drain-side metal semiconductor alloy portion 24d are thinned.

[0078] The anneal used to provide the thickened source-side transition metal-metal semiconductor alloy portion 26s', the thickened drain-side transition metal-metal semiconductor alloy portion 26d', and optionally the thickened transition metal-metal semiconductor alloy cap

27' can be performed at a temperature from 300°C to 500°C. The anneal used to provide the thickened source-side transition metal-metal semiconductor alloy portion 26s', the thickened drain-side transition metal-metal semiconductor alloy portion 26d', and optionally the thickened transition metal-metal semiconductor alloy cap 27' can be performed in an inert ambient such as, for example, helium, argon, neon and/or nitrogen. The anneal used to provide the thickened source-side transition metal-metal semiconductor alloy portion 26s', the thickened drain-side transition metal-metal semiconductor alloy portion 26d', and optionally the thickened transition metal-metal semiconductor alloy cap 27' can be performed utilizing one of the anneal processes mentioned above in forming the metal semiconductor alloy portions.

[0079] Referring now FIG. 7, there is illustrated the first exemplary semiconductor structure of FIG. 6 after removing remaining portions of the transition metal 26. The remaining portions of the transition metal 26 can be removed utilizing a selective etching process. As used herein, the term "selective" in reference to a material removal process denotes that the rate of material removal for a first material is greater than the rate of removal for at least another material of the structure to which the material removal process is being applied. In one embodiment, the selectivity may be greater than 10:1. In another embodiment, the selectivity may be greater than 100:1. An example of a selective etch that can be used to remove remaining portions of transition metal 26 comprises sulfuric acid and peroxide mixture OR aqua regia (HNO₃, HCl, and water).

[0080] FIG. 7 illustrates one semiconductor structure of the present application. As shown, the semiconductor substrate includes a semiconductor substrate 10 having a source region 22s and a drain region 22d located within a semiconductor material portion of the semiconductor substrate 10, wherein the source region 22s and the drain region 22d are spaced apart by a channel region 23. The semiconductor structure also includes a functional gate structure 18 located above the channel region 15, a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion 24s and a source-side transition metal-metal semiconductor alloy portion 26s or 26s' located on one side of the functional gate structure 18 and located atop the source region 22s, and a drain-side material stack of, from bottom to top, a drain-side metal

semiconductor alloy portion 24d and a drain-side transition metal-metal semiconductor alloy portion 26d or 26d' located on another side of the functional gate structure 18 and located atop the drain region 22d.

[0081] Referring now to FIG. 8, there is illustrated a second exemplary semiconductor structure including at least one sacrificial gate structure 54 formed on an upper surface of a semiconductor substrate 50 in accordance with another embodiment of the present application.

[0082] The semiconductor substrate 50 used in providing the second exemplary semiconductor structure shown in FIG. 8 is the same as semiconductor substrate 10 used in providing the first exemplary semiconductor structure shown in FIG. 1. Semiconductor substrate 50 may also be processed to include an isolation region 52. Isolation region 52 can be formed and include materials as described above in forming isolation regions 11 within semiconductor substrate 10.

[0083] The at least one sacrificial gate structure 54 is formed on an upper surface of semiconductor substrate 50. The term “sacrificial gate structure” is used throughout the present application to denote a material that serves as a placeholder structure for a functional gate structure to be subsequently formed.

[0084] The at least one sacrificial gate structure 54 can be formed by first providing a blanket layer of a sacrificial gate material on an upper surface of the semiconductor substrate 50. The blanket layer of sacrificial gate material can be formed, for example, by chemical vapor deposition or plasma enhanced chemical vapor deposition. The thickness of the blanket layer of sacrificial gate material can be from 50 nm to 300 nm, although lesser and greater thicknesses can also be employed. The blanket layer of sacrificial gate material can include any material that can be selectively removed from the structure during a subsequently performed etching process. In one embodiment, the blanket layer of sacrificial gate material may be composed of polysilicon. In another embodiment of the present application, the blanket layer of sacrificial gate material may be composed of a metal such as, for example, Al, W, or Cu. After providing

the blanket layer of sacrificial gate material, the blanket layer of sacrificial gate material can be patterned by lithography and etching so as to form the at least one sacrificial gate structure 54.

[0085] Referring now to FIG. 9, there is illustrated the second exemplary semiconductor structure of FIG. 8 after forming a spacer 56 on each vertical sidewall of the at least one sacrificial gate structure 54 and forming a source region 58s on one side of the at least one sacrificial gate structure 54 and a drain region 58d on another side of the at least one sacrificial gate structure 54. The semiconductor portion of the semiconductor substrate 50 that is positioned between the source region 58s and the drain region 58d is referred as a channel region 59.

[0086] The spacer 56 comprises one of the spacer materials mentioned above in forming spacer 22 to the first exemplary semiconductor structure. Spacer 56 can also be formed utilizing the technique mentioned above in forming spacer 22. The source region 58s and the drain region 58d can be formed by ion implantation as was also mentioned above in forming the source region 24s and the drain region 24d in the first exemplary semiconductor structure.

[0087] Referring now to FIG. 10, there is illustrated the second exemplary semiconductor structure of FIG. 9 after forming a dielectric material 60 and thereafter forming a gate cavity 62 by removing the at least one sacrificial gate structure 54. Dielectric material 60 has an upper surface that is coplanar with an upper surface of the at least one gate structure 54. As such, the upper surface of the at least one gate structure 54 is exposed after forming the dielectric material 60.

[0088] In some embodiments, the dielectric material 60 may be composed of, for example, silicon dioxide, undoped silicate glass (USG), fluorosilicate glass (FSG), borophosphosilicate glass (BPSG), a spin-on low-k dielectric layer, a chemical vapor deposition (CVD) low-k dielectric layer or any combination thereof. The term “low-k” as used throughout the present application denotes a dielectric material that has a dielectric constant of less than silicon dioxide. In another embodiment, a self-planarizing material such as a spin-on glass (SOG) or a spin-on

low-k dielectric material such as SiLKTM can be used as the dielectric material 60. The use of a self-planarizing dielectric material as dielectric material 60 may avoid the need to perform a subsequent planarizing step.

[0089] In one embodiment, the dielectric material 60 can be formed utilizing a deposition process including, for example, chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), evaporation or spin-on coating. In some embodiments, particularly when non-self-planarizing dielectric materials are used as dielectric material 60, a planarization process or an etch back process follows the deposition of the dielectric material. The thickness of the dielectric material 60 that can be employed in the present application may vary depending on the type of dielectric material employed as well as the method that was employed in forming the same. In one embodiment, the dielectric material 60 has a thickness from 80 nm to 500 nm. Other thicknesses that are greater or lesser than the range provided above can also be used for the dielectric material 60.

[0090] Each gate cavity 62 can be formed by removing each of the sacrificial gate material structures 54 from the structure. Each sacrificial gate material structure 54 can be removed by etching. In one example, a reactive ion etch can be used to removal each sacrificial gate material structure 54.

[0091] Referring now to FIG. 11, there is illustrated the second exemplary semiconductor structure of FIG. 10 after forming a gate dielectric portion 64 and a gate conductor portion 66 within the gate cavity 62. The gate dielectric portion 64 and the gate conductor portion 66 provide a functional gate structure 63 within the area previously occupied by the sacrificial gate structure 54.

[0092] In some embodiments and as shown in the drawing, the gate dielectric portion 64 is U-shaped having a bottommost portion in direct contact with an upper surface of the semiconductor substrate 50 and vertical portions that are located on exposed sidewalls of each spacer 56. Within each gate cavity 62, the gate dielectric portion 64 surrounds the gate conductor portion

64. In another embodiment, the gate dielectric portion 64 is not U-shaped and thus lacks the vertical portions mentioned. In such an embodiment, the gate conductor portion 66 that is formed atop the non-U-shaped gate dielectric fills the remaining portion of the gate cavity and has outermost edges that directly contact a sidewall surface of each spacer 56.

[0093] The gate dielectric portion 64 may comprise one of the gate dielectric materials mentioned above for gate dielectric material 14. Also, the gate dielectric portion 64 may be formed utilizing one of the deposition techniques or thermal growth techniques mentioned above in forming gate dielectric 14. The thickness of gate dielectric portion 64 is also with the range mentioned above for gate dielectric material 14.

[0094] The gate conductor portion 66 may comprise one of the metal gate conductor materials mentioned above for gate conductor 16. Also, the gate dielectric portion 66 may be formed utilizing one of the deposition techniques mentioned above in forming gate conductor 16. The thickness of gate dielectric portion 66 is also with the range mentioned above for gate conductor 16.

[0095] Referring now to FIG. 12, there is illustrated the second exemplary semiconductor structure of FIG. 11 after forming a dielectric cap material 61 on an upper surface of the dielectric material 60. The dielectric cap material 61 includes an insulator material that differs from the dielectric material 60. Examples of insulator materials that can be used as the dielectric cap material 61 include, for example, silicon dioxide, silicon nitride and silicon oxynitride. The dielectric cap material 61 can be formed by a deposition process including, for example, chemical vapor deposition, plasma enhanced chemical vapor deposition, evaporation and spin-on coating. The dielectric cap material 61 has a thickness from 1 nm to 20 nm. Other thicknesses can also be used for the thickness of the dielectric cap material 61.

[0096] Referring now to FIG. 13, there is illustrated the second exemplary semiconductor structure of FIG. 12 after forming a plurality of openings 65 through the dielectric cap material 61 and the dielectric material 60 exposing the source region 58s and the drain region 58d. Each

remaining portion of the dielectric cap material 60 is referred to herein as dielectric cap material portion 61c, while each remaining portion of the dielectric material 60 can be referred to herein as dielectric material portion 60'. The plurality of openings can be formed by lithography and etching. One or more etching processes can be used in forming the plurality of openings 65. The remaining portion of the dielectric cap portion 61c is present on a topmost surface of each functional gate structure 63. As is shown in the drawings of the present application, the dielectric cap portion 61c that is located on the topmost surface of each functional gate structure 63 extends beyond the vertical edges of the functional gate structure 63 and beyond an outermost edge of each spacer 56. Also, a portion of each dielectric cap portion 61c that is present on the topmost surface of the functional gate structure 63 is present on a dielectric material portion 60' that remains adjacent to the spacer 56. Further, the dielectric cap portion 61c and the dielectric material portion 60' that is present above the isolation regions 52 extend beyond the outer most edges of the isolation regions 52.

[0097] Referring now to FIG. 14, there is illustrated the second exemplary semiconductor structure of FIG. 13 after forming a source-side metal semiconductor alloy portion 68s on the source region 58s and a drain-side metal semiconductor alloy portion 68d on the drain region 58d. In this embodiment, no metal semiconductor alloy cap portion is formed atop the gate conductor portion 66.

[0098] The source-side metal semiconductor alloy portion 68s and the drain-side metal semiconductor alloy portion 68d can include one of the metal semiconductor alloys mentioned above for source-side metal semiconductor alloy portion 24s and drain-side metal semiconductor alloy portion 24d. Further, the source-side metal semiconductor alloy portion 68s and the drain-side metal semiconductor alloy portion 68d can be formed utilizing the processing steps mentioned above in forming source-side metal semiconductor alloy portion 24s and drain-side metal semiconductor alloy portion 24d.

[0099] Referring now to FIG. 15, there is illustrated the second exemplary semiconductor structure of FIG. 14 after deposition of a transition metal 70 in which a source-side transition

metal-metal semiconductor alloy portion 70s forms preferably *in-situ* on the source-side metal semiconductor alloy portion 68s and a drain-side transition metal-metal semiconductor alloy portion 70d forms preferably *in-situ* on the drain-side metal semiconductor alloy portion 68d.

[0100] The transition metal 70, source-side transition metal-metal semiconductor alloy portion 70s and drain-side transition metal-metal semiconductor alloy portion 70d are formed as described above in forming transition metal 26, source-side transition metal-metal semiconductor alloy portion 26s and drain-side transition metal-metal semiconductor alloy portion 26d. Also, the transition metal 70, source-side transition metal-metal semiconductor alloy portion 70s and drain-side transition metal-metal semiconductor alloy portion 70d can include one of the transition metal mentioned above in the transition metal 26, source-side transition metal-metal semiconductor alloy portion 26s and drain-side transition metal-metal semiconductor alloy portion 26d. In this embodiment of the present application, portions of the transition material 26 are present on the vertical sidewalls of each dielectric material portion 60' and each dielectric cap material portion 61c. Also, a portion of the transition metal 26 extends onto an upper surface of each dielectric cap material portion 61c.

[0101] Referring now to FIG. 16, there is illustrated the second exemplary semiconductor structure of FIG. 15 after annealing. This anneal is optional and thus not be employed in all instances. The anneal step forms a thickened source-side transition metal-metal semiconductor alloy portion 70s' and a thickened drain-side transition metal-metal semiconductor alloy portion 70d'. The anneal used in this embodiment of the present application is the same as that described above in forming the first exemplary semiconductor structure shown in FIG. 6.

[0102] Referring now to FIG. 17, there is illustrated the second exemplary semiconductor structure of FIG. 16 after removing remaining portions of the transition metal 70. The remaining portions of the transition metal 70 can be removed utilizing a selective etching process such as described above for removing remaining portions of transition metal 26 from the first exemplary semiconductor structure.

[0103] FIG. 17 illustrates another semiconductor structure of the present application. As shown the semiconductor substrate includes a semiconductor substrate 50 having a source region 58s and a drain region 58d located within a semiconductor material portion of the semiconductor substrate 50, wherein the source region 58s and the drain region 58d are spaced apart by a channel region 59. The semiconductor structure also includes a functional gate structure 63 located above the channel region 59, a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion 68s and a source-side transition metal-metal semiconductor alloy portion 70s or 70s' located on one side of the functional gate structure 63 and located atop the source region 58s, and a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion 68d and a drain-side transition metal-metal semiconductor alloy portion 70d or 70d' located on another side of the functional gate structure 63 and located atop the drain region 58d.

[0104] A conductive material (not shown) can then be deposited filling each opening. The conductive material that is deposited may include for example, at least one of copper, tungsten, and aluminum. In one embodiment, the conductive material that fills each opening comprises copper or a copper alloy such as, for example, a copper-aluminum alloy. The conductive material may be formed by a deposition process including chemical vapor deposition (CVD), plasma enhanced chemical vapor deposition (PECVD), sputtering, or chemical solution deposition. Alternatively, a plating process that fills each contact opening from the bottom upwards can be used. After depositing the conductive material, a planarization process such as, for example, chemical mechanical polishing (CMP) and/or grinding, can be employed to remove portions of the conductive material that extends above the mouth of each opening. In some embodiments, the planarization step can stop on an upper surface of each dielectric cap portion 51c. In another embodiment, the planarization step can remove each dielectric cap portion 51c from the structure.

[0105] Referring to FIG. 18, there is illustrated the second exemplary semiconductor structure of FIG. 9 after forming a source-side metal semiconductor alloy portion 68s on the source region 58s and the drain-side metal semiconductor alloy portion 68d of the drain region 58d in

accordance with an alternative embodiment of the present application. In this embodiment, no metal semiconductor alloy cap portion is formed atop the gate conductor portion 66.

[0106] The source-side metal semiconductor alloy portion 68s and the drain-side metal semiconductor alloy portion 68d can include one of the metal semiconductor alloys mentioned above for source-side metal semiconductor alloy portion 24s and drain-side metal semiconductor alloy portion 24d. Further, the source-side metal semiconductor alloy portion 68s and the drain-side metal semiconductor alloy portion 68d can be formed utilizing the processing steps mentioned above in forming source-side metal semiconductor alloy portion 24s and drain-side metal semiconductor alloy portion 24d.

[0107] Referring now to FIG. 19, there is illustrated the second exemplary semiconductor structure of FIG. 18 after deposition of a transition metal 70 in which a source-side transition metal-metal semiconductor alloy portion 70s forms preferably *in-situ* on the source-side metal semiconductor alloy portion 68s and a drain-side transition metal-metal semiconductor alloy portion 70d forms preferably *in-situ* on the drain-side metal semiconductor alloy portion 68d.

[0108] The transition metal 70, source-side transition metal-metal semiconductor alloy portion 70s and drain-side transition metal-metal semiconductor alloy portion 70d are formed as described above in forming transition metal 26, source-side transition metal-metal semiconductor alloy portion 26s and drain-side transition metal-metal semiconductor alloy portion 26d. Also, the transition metal 70, source-side transition metal-metal semiconductor alloy portion 70s and drain-side transition metal-metal semiconductor alloy portion 70d can include one of the transition metal mentioned above in the transition metal 26, source-side transition metal-metal semiconductor alloy portion 26s and drain-side transition metal-metal semiconductor alloy portion 26d. The source-side transition metal-metal semiconductor alloy portion 70s and drain-side transition metal-metal semiconductor alloy portion 70d each contain up to 70% of transition metal.

[0109] Referring now to FIG. 20, there is illustrated the second exemplary semiconductor structure of FIG. 19 after annealing. This anneal is optional and thus not need be performed in all instances. The anneal step forms a thickened source-side transition metal-metal semiconductor alloy portion 70s' and a thickened drain-side transition metal-metal semiconductor alloy portion 70d'. The anneal used in this embodiment of the present application is the same as that described above in forming the first exemplary semiconductor structure shown in FIG. 6.

[0110] Referring now to FIG. 21, there is illustrated the second exemplary semiconductor structure of FIG. 20 after removing remaining portions of the transition metal 70. The remaining portions of the transition metal 70 can be removed utilizing a selective etching process such as described above for removing remaining portions of transition metal 26 from the first exemplary semiconductor structure.

[0111] Referring now to FIG. 22, there is illustrated the second exemplary semiconductor structure of FIG. 21 after forming a dielectric material 60. The dielectric material 60 that can be employed in this alternative embodiment of the present application includes one of the dielectric materials mentioned above for dielectric material 60. Also, the dielectric material 60 that can be employed in this alternative embodiment of the present application formed utilizing one of the techniques mentioned above for forming dielectric material 60.

[0112] Referring now to FIG. 22, there is illustrated the second exemplary semiconductor structure of FIG. 21 after forming a gate cavity by removing the at least one sacrificial gate structure 54, and forming a functional gate structure including a gate dielectric portion 64 and a gate conductor portion 66 within the gate cavity.

[0113] Each gate cavity can be formed utilizing the technique mentioned above in forming the gate cavity in the second exemplary structure of FIG. 10. Also, the description concerning the gate dielectric portion 64 and gate conductor portion 66 used in providing the second exemplary structure shown in FIG. 11 can also be used in providing the gate dielectric portion 64 and gate

conductor portion 66 within the gate cavity. In this embodiment, the gate dielectric portion 64 is not U-shaped. Elements 64 and 66 collectively made by referred to as a functional gate structure 63.

[0114] FIG. 22 illustrates another semiconductor structure of the present application. As shown, the semiconductor substrate includes a semiconductor substrate 50 having a source region 58s and a drain region 58d located within a semiconductor material portion of the semiconductor substrate 50, wherein the source region 58s and the drain region 58d are spaced apart by a channel region 59. The semiconductor structure also includes a functional gate structure 63 located above the channel region 59, a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion 68s and a source-side transition metal-metal semiconductor alloy portion 70s or 70s' located on one side of the functional gate structure 63 and located atop the source region 58s, and a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion 68d and a drain-side transition metal-metal semiconductor alloy portion 70d or 70d' located on another side of the functional gate structure 63 and located atop the drain region 58d.

[0115] The method of the present application for forming the source-side material stack on a source region and the drain-side material stack on a drain region is not limited to the specific examples mentioned above. Instead, the method of the present application for forming the source-side material stack on a source region and the drain-side material stack on a drain region can be used in any other processing schemes including, for example, formation of FinFET and semiconductor nanowire devices. In addition, the method of the present application can be used in forming contacts to any other type of semiconductor device such as, for example, a hetero-bipolar junction transistor (HBT). Also, the deposition of the transition metal onto a metallic surface of a gate conductor portion may cause some of the transition metal to diffuse into the metallic surface of the gate conductor portion.

[0116] While the present application has been particularly shown and described with respect to various embodiments thereof, it will be understood by those skilled in the art that the foregoing

and other changes in forms and details may be made without departing from the spirit and scope of the present application. It is therefore intended that the present application not be limited to the exact forms and details described and illustrated, but fall within the scope of the appended claims.

CLAIMS

What is claimed is:

1. A semiconductor structure comprising:

a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of said semiconductor substrate, wherein said source region and said drain region are spaced apart by a channel region;

a functional gate structure located above said channel region;

a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition-metal metal semiconductor alloy portion located on one side of said functional gate structure and located atop said source region; and

a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of said functional gate structure and located atop said drain region.

2. The semiconductor structure of Claim 1, wherein outermost edges of said source-side metal semiconductor alloy portion and outermost edges of said source-side transition metal-metal semiconductor alloy portion are vertically coincident to each other, and wherein outermost edges of said drain-side metal semiconductor alloy portion and outermost edges of said drain-side transition metal-metal semiconductor alloy portion are vertically coincident to each other.

3. The semiconductor structure of Claim 1, wherein said source-side transition metal-metal semiconductor alloy portion is more etch resistant than said source-side metal semiconductor alloy portion, and said drain-side transition metal-metal semiconductor alloy portion is more etch resistant than said drain-side metal semiconductor alloy portion.

4. The semiconductor structure of Claim 1, wherein said transition metal of said source-side transition metal-metal semiconductor alloy portion comprises a different metal than said metal of said source-side metal semiconductor alloy portion, and said transition metal of said drain-side

transition metal-metal semiconductor alloy portion comprises a different metal than said metal of said drain-side metal semiconductor alloy portion.

5. The semiconductor structure of Claim 4, wherein said semiconductor substrate comprises silicon, said source-side metal semiconductor alloy portion comprises nickel silicide or platinum-nickel silicide, said source-side transition-metal metal semiconductor alloy portion comprises titanium-nickel or titanium-platinum-nickel silicide, said drain-side metal semiconductor alloy portion comprises nickel silicide and said drain-side transition metal-metal semiconductor alloy comprises titanium-nickel silicide or titanium-platinum-nickel silicide.

6. The semiconductor structure of Claim 1, wherein said functional gate structure comprises a dielectric material portion and a gate conductor portion, and said dielectric material portion is U-shaped.

7. The semiconductor structure of Claim 1, further comprising a dielectric cap material portion located on a topmost surface of said functional gate structure.

8. The semiconductor structure of Claim 6, wherein said gate conductor portion includes a transition metal-metal semiconductor alloy cap portion located on a surface of a metal semiconductor alloy cap portion.

9. The semiconductor structure of Claim 1, wherein said semiconductor substrate is a III-V compound semiconductor material.

10. A method of forming a semiconductor structure comprising:

providing a source region and a drain region at a footprint of a functional gate structure;
forming a source-side metal semiconductor alloy portion on said source region, and a drain-side metal semiconductor alloy portion on said drain region;
depositing a transition metal on at least said source-side metal semiconductor alloy portion and said drain-side metal semiconductor alloy portion, wherein during said depositing

said transition metal reacts *in-situ* with said source-side metal semiconductor alloy portion forming a source-side transition metal-metal semiconductor alloy portion on a surface of said source-side metal semiconductor alloy portion, and said transition metal reacts *in-situ* with said drain-side metal semiconductor alloy portion forming a drain-side transition metal-metal semiconductor alloy portion on a surface of said drain-side metal semiconductor alloy portion; and

removing remaining portions of said transition metal.

11. The method of Claim 10, wherein prior to removing said remaining portions of said transition metal, an anneal step is performed to thicken both the source-side transition metal-metal semiconductor alloy portion and said drain-side transition metal semiconductor alloy portion.

12. The method of Claim 10, wherein outermost edges of said source-side metal semiconductor alloy portion and outermost edges of said source-side transition metal-metal semiconductor alloy portion are vertically coincident to each other, and wherein outermost edges of said drain-side metal semiconductor alloy portion and outermost edges of said drain-side transition metal-metal semiconductor alloy portion are vertically coincident to each other.

13. The method of Claim 10, wherein said transition metal comprises a different metal than said metal of said source-side metal semiconductor alloy portion and of said drain-side metal semiconductor alloy portion.

14. The method of Claim 10, wherein said semiconductor substrate comprises silicon, said source-side metal semiconductor alloy portion comprises nickel silicide, said drain-side metal semiconductor alloy portion comprises nickel silicide and said transition metal comprises titanium.

15. The method of Claim 10, wherein said depositing said transition metal is performed by sputtering at a temperature from room temperature up to, and including 500°C.

16. A method of forming a semiconductor substrate comprising:

- providing a source region and a drain region at a footprint of a sacrificial gate structure;
- forming a source-side metal semiconductor alloy portion on said source region, and a drain-side metal semiconductor alloy portion on said drain region;
- depositing a transition metal on at least said source-side metal semiconductor alloy portion and said drain-side metal semiconductor alloy portion, wherein during said depositing said transition metal reacts *in-situ* with said source-side metal semiconductor alloy portion forming a source-side transition metal-metal semiconductor alloy portion on a surface of said source-side metal semiconductor alloy portion, and said transition metal reacts *in-situ* with said drain-side metal semiconductor alloy portion forming a drain-side transition metal-metal semiconductor alloy portion on a surface of said drain-side metal semiconductor alloy portion;
- and
- removing remaining portions of said transition metal, wherein said sacrificial gate structure is replaced with a functional gate structure prior to forming the source-side metal semiconductor alloy portion on the source region, and the drain-side metal semiconductor alloy portion on the drain region, or after removing the transition metal.

17. The method of Claim 16, wherein prior to removing said remaining portions of said transition metal, an anneal step is performed to thicken both the source-side transition metal-metal semiconductor alloy portion and said drain-side transition metal semiconductor alloy portion.

18. The method of Claim 16, wherein outermost edges of said source-side metal semiconductor alloy portion and outermost edges of said source-side transition metal-metal semiconductor alloy portion are vertically coincident to each other, and wherein outermost edges of said drain-side metal semiconductor alloy portion and outermost edges of said drain-side transition metal-metal semiconductor alloy portion are vertically coincident to each other.

19. The method of Claim 16, wherein said transition metal comprises a different metal than said metal of said source-side metal semiconductor alloy portion and of said drain-side metal semiconductor alloy portion.

20. The method of Claim 16, wherein said semiconductor substrate comprises silicon, said source-side metal semiconductor alloy portion comprises nickel silicide or platinum-nickel silicide, said drain-side metal semiconductor alloy portion comprises nickel silicide or platinum-nickel silicide, and said transition metal comprises titanium.

ABSTRACT

Metal semiconductor alloy contacts are provided on each of a source region and a drain region which are present in a semiconductor substrate. A transition metal is then deposited on each of the metal semiconductor alloy contacts, and during the deposition of the transition metal, the deposited transition metal reacts preferably, but not necessarily always, *in-situ* with a portion of each the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact. Each transition metal-metal semiconductor alloy liner that is provided has outer edges that are vertically coincident with outer edges of each metal semiconductor alloy contact. The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process.

PATENT ASSIGNMENT COVER SHEET

Electronic Version v1.1
Stylesheet Version v1.2

EPAS ID: PAT2722041

SUBMISSION TYPE:	NEW ASSIGNMENT												
NATURE OF CONVEYANCE:	ASSIGNMENT												
CONVEYING PARTY DATA													
<table border="1"><thead><tr><th>Name</th><th>Execution Date</th></tr></thead><tbody><tr><td>NICOLAS BREIL</td><td>10/24/2013</td></tr><tr><td>CHRISTIAN LAVOIE</td><td>10/25/2013</td></tr><tr><td>AHMET S OZCAN</td><td>10/25/2013</td></tr><tr><td>KATHRYN T SCHONENBERG</td><td>10/25/2013</td></tr><tr><td>JIAN YU</td><td>10/28/2013</td></tr></tbody></table>		Name	Execution Date	NICOLAS BREIL	10/24/2013	CHRISTIAN LAVOIE	10/25/2013	AHMET S OZCAN	10/25/2013	KATHRYN T SCHONENBERG	10/25/2013	JIAN YU	10/28/2013
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JIAN YU	10/28/2013												
RECEIVING PARTY DATA													
Name:	INTERNATIONAL BUSINESS MACHINES CORPORATION												
Street Address:	NEW ORCHARD ROAD												
City:	ARMONK												
State/Country:	NEW YORK												
Postal Code:	10504												
PROPERTY NUMBERS Total: 1													
<table border="1"><thead><tr><th>Property Type</th><th>Number</th></tr></thead><tbody><tr><td>Application Number:</td><td>14177481</td></tr></tbody></table>		Property Type	Number	Application Number:	14177481								
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Fax Number:	(845)892-6363												
Phone:	516-742-4343												
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<i>Correspondence will be sent via US Mail when the email attempt is unsuccessful.</i>													
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NAME OF SUBMITTER:	CATHERINE IVERS												

Signature:	/Catherine Ivers/
Date:	02/11/2014
	This document serves as an Oath/Declaration (37 CFR 1.63).
Total Attachments: 6 source=FIS920130015US1_Declaration_Assignment#page1.tif source=FIS920130015US1_Declaration_Assignment#page2.tif source=FIS920130015US1_Declaration_Assignment#page3.tif source=FIS920130015US1_Declaration_Assignment#page4.tif source=FIS920130015US1_Declaration_Assignment#page5.tif source=FIS920130015US1_Declaration_Assignment#page6.tif	

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY
CONTACTS**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☐ United States application or PCT international application number _____
filed on _____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

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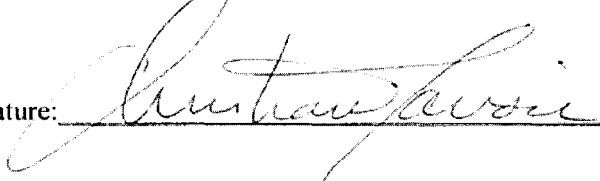
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I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Nicolas Breil**

Signature: _____ Date: _____

(2) Legal Name of Inventor: **Christian Lavoie**

Signature:  _____ Date: 10/25/2013

(3) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Jian Yu**

Signature: _____ Date: _____

IBM DOCKET NUMBER: FIS920130015US1

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As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☐ United States application or PCT international application number
filed on _____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the
application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information
known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries
(herein referred to as the "Invention") disclosed in the above-identified patent application and further
identified by the IBM Docket Number provided above in the header of this document:

Whereas, International Business Machines Corporation, a corporation of New York having a place of
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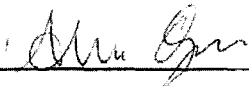
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Signature: _____ Date: _____

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Signature:  Date: 10/25/2013

(4) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature: _____ Date: _____

(5) Legal Name of Inventor: **Jian Yu**

Signature: _____ Date: _____

**DECLARATION (37 C.F.R. 1.63) FOR UTILITY PATENT APPLICATION USING AN
APPLICATION DATA SHEET (37 C.F.R. 1.76) AND ASSIGNMENT**

Title of Invention: **SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY
CONTACTS**

As a below named inventor, I hereby declare that:

This declaration is directed to the attached application, or (if following box is checked):

☐ United States application or PCT international application number _____
filed on _____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I have reviewed and understand the contents of the application, including the claims.

I am aware of the duty to disclose to the United States Patent and Trademark Office all information known to me to be material to patentability as defined in 37 CFR Section 1.56.

Whereas, the undersigned inventor(s) has/have made certain inventions, improvements, and discoveries (herein referred to as the "Invention") disclosed in the above-identified patent application and further identified by the IBM Docket Number provided above in the header of this document;

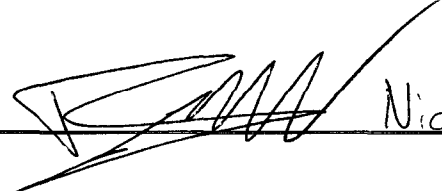
Whereas, International Business Machines Corporation, a corporation of New York having a place of business at Armonk, New York (herein referred to as "IBM"), desires to acquire, and each undersigned inventor desires to grant to IBM, the entire worldwide right, title, and interest in and to the Invention and in and to any and all patent applications and patents directed thereto;

Now, therefore, for good and valuable consideration, the receipt and sufficiency thereof being hereby acknowledged, each undersigned inventor ("ASSIGNOR") hereby sells or has sold, assigns or has assigned, and otherwise transfers or has transferred to IBM (the "ASSIGNEE"), its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the Invention, the above-identified United States patent application, and any and all other patent applications and patents for the Invention which may be applied for or granted therefor in the United States and in all foreign countries and jurisdictions, including all divisions, continuations, reissues, reexaminations, renewals, extensions, counterparts, substitutes, and extensions thereof, and all rights of priority resulting from the filing of such applications and granting of such patents. In addition, each undersigned inventor hereby authorizes and requests the Director of the United States Patent and Trademark Office to issue any United States Patent, and foreign patent authorities to issue any foreign patent, granted for the Invention, to IBM, its successors, legal representatives, and assigns, the entire worldwide right, title, and interest in and to the same to be held and enjoyed by IBM, its successors, legal representatives, and assigns to the full end of the terms for which any and all such patents may be granted, as fully and entirely as would have been held and enjoyed by the undersigned had this Assignment not been made; and each undersigned inventor agrees to execute any and all documents and instruments and perform all lawful acts reasonably related to recording this Assignment or perfecting title to the Invention and all related patents and applications, in IBM, its successors, legal representatives, and assigns, whenever requested by IBM, its successors, legal representatives, or assigns.

Each undersigned inventor acknowledges their prior and ongoing obligations to sell, assign, and transfer the rights under this Assignment to IBM and is unaware of any reason why they may not have the full and unencumbered right to sell, assign, and transfer the rights hereby sold, assigned, and transferred, and has not executed, and will not execute, any document or instrument in conflict herewith. Each undersigned inventor also hereby grants IBM, its successors, legal representatives, and assigns, the right to insert in this Assignment any further identification (including, but not limited to, patent Application Number) which may be necessary or desirable for recordation of this Assignment. This Assignment is governed by the substantive laws of the State of New York, and any disputes will be resolved in a New York state court or federal court sited in New York.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

(1) Legal Name of Inventor: **Nicolas Breil**

Signature:  Nicolas Breil Date: 10/24/2013

(2) Legal Name of Inventor: **Christian Lavoie**

Signature: _____ Date: _____

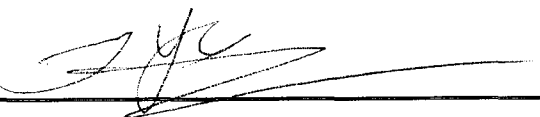
(3) Legal Name of Inventor: **Ahmet S. Ozcan**

Signature: _____ Date: _____

(4) Legal Name of Inventor: **Kathryn T. Schonenberg**

Signature:  _____ Date: 10/25/2013

(5) Legal Name of Inventor: **Jian Yu**

Signature:  _____ Date: 10/28/2013



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/177,481	02/11/2014	2812	1600	FIS920130015US1(29690)	20	3

CONFIRMATION NO. 5539

45600
SCULLY, SCOTT, MURPHY & PRESSER, P.C.
400 GARDEN CITY PLAZA
Suite 300
GARDEN CITY, NY 11530

FILING RECEIPT



Date Mailed: 02/27/2014

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Nicolas Breil, Grenoble, FRANCE;
Christian Lavoie, Ossining, NY;
Ahmet S. Ozcan, Pleasantville, NY;
Kathryn T. Schonenberg, Wappingers Falls, NY;
Jian Yu, Danbury, CT;

Applicant(s)

International Business Machines Corporation, Armonk, NY

Assignment For Published Patent Application

International Business Machines Corporation, Armonk, NY

Power of Attorney: None

Domestic Applications for which benefit is claimed - None.

A proper domestic benefit claim must be provided in an Application Data Sheet in order to constitute a claim for domestic benefit. See 37 CFR 1.76 and 1.78.

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access - A proper **Authorization to Permit Access to Application by Participating Offices** (PTO/SB/39 or its equivalent) has been received by the USPTO.

If Required, Foreign Filing License Granted: 02/24/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/177,481**

Projected Publication Date: 08/13/2015

Non-Publication Request: No

Early Publication Request: No

Title

SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS

Preliminary Class

438

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/177,481

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	280
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	600
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	720
TOTAL CLAIMS (37 CFR 1.16(j))	20	minus 20 = *			OR	x 80 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3	minus 3 = *				x 420 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
			TOTAL			TOTAL	1600

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/177,481	02/11/2014	Nicolas Breil	FIS920130015US1(29690)

CONFIRMATION NO. 5539
IMPROPER CPOA LETTER

45600
SCULLY, SCOTT, MURPHY & PRESSER, P.C.
400 GARDEN CITY PLAZA
Suite 300
GARDEN CITY, NY 11530



Date Mailed: 02/27/2014

NOTICE REGARDING POWER OF ATTORNEY

This is in response to the power of attorney filed 02/11/2014. The power of attorney in this application is not accepted for the reason(s) listed below:

- The power of attorney has not been accepted because the party who is giving power has not been identified. Power of attorney may only be signed by the applicant for patent (37 CFR 1.42) or the patent owner. A party who is not the applicant must become the applicant in accordance with 37 CFR 1.46(c) and appoint any power of attorney in compliance with 37 CFR 3.71 and 3.73. For a reissue application, reexamination proceeding, or supplemental examination proceeding, a patent owner who was not the applicant under 37 CFR 1.46 must appoint any power of attorney in compliance with 37 CFR 3.71 and 3.73. See 37 CFR 1.32(b)(4).

/yhailu/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	14177481
Filing Date	02-11-2014
First Named Inventor	Nicolas Breil
Title	SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS
Art Unit	2812
Examiner Name	Unassigned
Attorney Docket Number	FIS920130015US1(29690)

SIGNATURE of Applicant or Patent Practitioner

Signature	/Leslie S. Szivos/	Date (Optional)	10-02-2014
Name	Leslie S. Szivos, Ph.D.	Registration Number	39394
Title (if Applicant is a juristic entity)	Patent Practitioner for Applicant		
Applicant Name (if Applicant is a juristic entity)	International Business Machines Corporation		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.



*Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number	Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)



I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

45600

OR



I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:



The address associated with the above-mentioned Customer Number

OR



The address associated with Customer Number:

OR

Firm or
Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

International Business Machines Corporation



Inventor or Joint Inventor (title not required below)



Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)



Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)



Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Pryor A. Garnett

Date (Optional)

11/07/13

Name

Pryor A. Garnett, Reg. # 32,136

Title

Senior Counsel

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.



Total of 1 (one) forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Acknowledgement Receipt

EFS ID:	20302322
Application Number:	14177481
International Application Number:	
Confirmation Number:	5539
Title of Invention:	SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS
First Named Inventor/Applicant Name:	Nicolas Breil
Customer Number:	45600
Filer:	Leslie S. Szivos/Tiffani Moran
Filer Authorized By:	Leslie S. Szivos
Attorney Docket Number:	FIS920130015US1(29690)
Receipt Date:	02-OCT-2014
Filing Date:	11-FEB-2014
Time Stamp:	16:23:53
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	FIS920130015US1_PowerOfAttorney2.pdf	210920 ff21b186ef2cfe05d48c2efafda07e7762beb713	no	2

Warnings:

Information:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/177,481	02/11/2014	Nicolas Breil	FIS920130015US1(29690)

CONFIRMATION NO. 5539

POA ACCEPTANCE LETTER

45600
SCULLY, SCOTT, MURPHY & PRESSER, P.C.
400 GARDEN CITY PLAZA
Suite 300
GARDEN CITY, NY 11530



Date Mailed: 10/09/2014

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 10/02/2014.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/cnguyen/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/177,481	02/11/2014	Nicolas Breil	FIS920130015US1(29690)	5539

45600 7590 03/20/2015
SCULLY, SCOTT, MURPHY & PRESSER, P.C.
400 GARDEN CITY PLAZA
Suite 300
GARDEN CITY, NY 11530

EXAMINER

PHAM, HOAI V

ART UNIT	PAPER NUMBER
----------	--------------

2892

NOTIFICATION DATE	DELIVERY MODE
-------------------	---------------

03/20/2015

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

IBMPAIRENotify@ssmp.com

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Election/Restrictions

1. Restriction to one of the following inventions is required under 35 U.S.C. 121:
 - I. Claims 1-9, drawn to a semiconductor device, classified in CPC H01L 29/49.
 - II. Claims 10-20, drawn to a method of making a semiconductor device, classified in CPC H01L 21/8238.
2. The inventions are distinct, each from the other because of the following reasons:

Inventions I and II are related as process of making and product made. The inventions are distinct if either or both of the following can be shown: (1) that the process as claimed can be used to make another and materially different product or (2) that the product as claimed can be made by another and materially different process (MPEP § 806.05(f)). In the instant case, the product as claimed can be made by another and materially different process such as the transition metal can be formed by ex-situ instead of in-situ.
3. Restriction for examination purposes as indicated is proper because all these inventions listed in this action are independent or distinct for the reasons given above and there would be a serious search and examination burden if restriction were not required because one or more of the following reasons apply:

- (a) the inventions have acquired a separate status in the art in view of their different classification;
- (b) the inventions have acquired a separate status in the art due to their recognized divergent subject matter;
- (c) the inventions require a different field of search (for example, searching different classes/subclasses or electronic resources, or employing different search queries);
- (d) the prior art applicable to one invention would not likely be applicable to another invention;
- (e) the inventions are likely to raise different non-prior art issues under 35 U.S.C. 101 and/or 35 U.S.C. 112, first paragraph.

Applicant is advised that the reply to this requirement to be complete must include (i) an election of a invention to be examined even though the requirement may be traversed (37 CFR 1.143) **and (ii) identification of the claims encompassing the elected invention.**

4. The election of an invention may be made with or without traverse. To reserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the restriction requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after

Art Unit: 2892

the election, applicant must indicate which of these claims are readable on the elected invention.

5. If claims are added after the election, applicant must indicate which of these claims are readable upon the elected invention.

6. Should applicant traverse on the ground that the inventions are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing the inventions to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the inventions unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103(a) of the other invention.

7. This application contains claims directed to the following patentably distinct species:

Species 1 of figures 1-7.

Species 2 of figures 8-17.

Species 3 of figures 18-23.

8. The species are independent or distinct because claims to the different species recite the mutually exclusive characteristics of such species. In addition, these species are not obvious variants of each other based on the current record.

9. Applicant is required under 35 U.S.C. 121 to elect a single disclosed species for prosecution on the merits to which the claims shall be restricted if no generic claim is finally held to be allowable.

10. There is an examination and search burden for these patentably distinct species due to their mutually exclusive characteristics. The species require a different field of search (e.g., searching different classes/subclasses or electronic resources, or employing different search queries); and/or the prior art applicable to one species would not likely be applicable to another species; and/or the species are likely to raise different non-prior art issues under 35 U.S.C. 101 and/or 35 U.S.C. 112, first paragraph.

11. **Applicant is advised that the reply to this requirement to be complete must include (i) an election of a species to be examined** even though the requirement may be traversed (37 CFR 1.143) **and (ii) identification of the claims encompassing the elected species**, including any claims subsequently added. An argument that a claim is allowable or that all claims are generic is considered nonresponsive unless accompanied by an election.

12. The election of the species may be made with or without traverse. To preserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the election of species requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable on the elected species.

13. Should applicant traverse on the ground that the species are not patentably distinct, applicant should submit evidence or identify such evidence now of record

Art Unit: 2892

showing the species to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the species unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103(a) of the other species.

14. Upon the allowance of a generic claim, applicant will be entitled to consideration of claims to additional species which depend from or otherwise require all the limitations of an allowable generic claim as provided by 37 CFR 1.141.

15. The examiner has required restriction between product and process claims.

Where applicant elects claims directed to the product, and the product claims are subsequently found allowable, withdrawn process claims that depend from or otherwise require all the limitations of the allowable product claim will be considered for rejoinder.

All claims directed to a nonelected process invention must require all the limitations of an allowable product claim for that process invention to be rejoined.

16. In the event of rejoinder, the requirement for restriction between the product claims and the rejoined process claims will be withdrawn, and the rejoined process claims will be fully examined for patentability in accordance with 37 CFR 1.104. Thus, to be allowable, the rejoined claims must meet all criteria for patentability including the requirements of 35 U.S.C. 101, 102, 103 and 112. Until all claims to the elected product are found allowable, an otherwise proper restriction requirement between product claims and process claims may be maintained. Withdrawn process claims that are not commensurate in scope with an allowable product claim will not be rejoined. See MPEP § 821.04(b). Additionally, in order to retain the right to rejoinder in accordance with the

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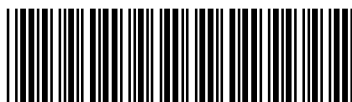
above policy, applicant is advised that the process claims should be amended during prosecution to require the limitations of the product claims. **Failure to do so may result in a loss of the right to rejoinder.** Further, note that the prohibition against double patenting rejections of 35 U.S.C. 121 does not apply where the restriction requirement is withdrawn by the examiner before the patent issues. See MPEP § 804.01.

17. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Hoai v. Pham whose telephone number is 571-272-1715. The examiner can normally be reached on M-F.

18. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Thao Xuan Le can be reached on 571-272-1708. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

19. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HOAI V PHAM/
Primary Examiner, Art Unit 2892

<i>Index of Claims</i> 	Application/Control No. 14177481	Applicant(s)/Patent Under Reexamination BREIL ET AL.
	Examiner HOAI V PHAM	Art Unit 2892

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	03/16/2015							
	1	÷							
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	3	÷							
	4	÷							
	5	÷							
	6	÷							
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	17	÷							
	18	÷							
	19	÷							
	20	÷							

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Nicolas Breil
Application No.: 14/177,481
Filed: February 11, 2014
Art Unit: 2892
Examiner: Hoai V Pham
Confirmation No.: 5539
Docket: FIS920130015US1(29690)
Dated: April 15, 2015
For: SELF-ALIGNED LINER FORMED ON METAL
SEMICONDUCTOR ALLOY CONTACTS

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

RESPONSE UNDER 37 C.F.R. §§ 1.111 AND 1.143

In response to the Office Action dated March 20, 2015, Applicant submits the following remarks for entry of record in the above identified patent application.

Remarks/Arguments begin on page 2 of this paper.

CERTIFICATE OF ELECTRONIC FILING

I hereby certify that this correspondence is being deposited with the United States Patent & Trademark Office via Electronic Filing through the United States Patent and Trademark Office e-business website on the date set forth below.

Dated: April 15, 2015



Leslie S. Szivos, Ph.D

REMARKS

Favorable consideration and allowance of the claims of the present application are respectfully requested.

In the present Official Action, Claims 1-20, which are present in the instant application, have been subjected to a restriction under 35 U.S.C. § 121 as follows:

- I. Claims 1-9, drawn to a semiconductor device, classified in CPC H01L 29/49.
- II. Claims 10-20, drawn to a method of making a semiconductor device, classified in CPC H01L 21/8238.

In order to be fully responsive to the restriction requirement imposed in the outstanding Official Action, Applicant elects, without traverse, to prosecute the subject matter of Group I, i.e., Claims 1-9, and reserves the right to file one or more divisional applications directed to the non-elected subject matter in this application, i.e., Claims 10-20.

In addition to the above Restriction Requirement, the Examiner also alleges that the application contains claims directed to the following patentably distinct species:

Species 1 of FIGS. 1-7.

Species 2 of FIGS. 8-17.

Species 3 of FIGS. 18-23.

In view of the above, Applicant elects, without traverse, Species 2, for continued prosecution. Applicant observes that Claims 1-6 and 8-9 read on Species 2. Applicant reserves the right to have the non-elected species rejoined in this application upon allowance of any generic claim. Presently, Claims 1-5 and 9 are generic.

Wherefore, consideration and allowance of the claims of the present application are respectfully requested.

Respectfully submitted,

A handwritten signature in dark ink, appearing to read 'Leslie S. Szivos', with a long horizontal flourish extending to the right.

Leslie S. Szivos, Ph.D.
Registration No. 39,394

Scully, Scott, Murphy & Presser, P.C.
400 Garden City Plaza - Suite 300
Garden City, New York 11530
(516) 742-4343
LSS:ch

Electronic Acknowledgement Receipt

EFS ID:	22067005
Application Number:	14177481
International Application Number:	
Confirmation Number:	5539
Title of Invention:	SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS
First Named Inventor/Applicant Name:	Nicolas Breil
Customer Number:	45600
Filer:	Leslie S. Szivos/Candice Hudson
Filer Authorized By:	Leslie S. Szivos
Attorney Docket Number:	FIS920130015US1(29690)
Receipt Date:	15-APR-2015
Filing Date:	11-FEB-2014
Time Stamp:	14:33:36
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		FIS920130015US1_Responseto RestrictionRequirement.pdf	172344 8f8ea404c91989cc89cb1fc980b68a6509993f06	yes	3

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Response to Election / Restriction Filed	1	1
	Applicant Arguments/Remarks Made in an Amendment	2	3
Warnings:			
Information:			
Total Files Size (in bytes):		172344	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

45600 7590 05/11/2015
SCULLY, SCOTT, MURPHY & PRESSER, P.C.
400 GARDEN CITY PLAZA
Suite 300
GARDEN CITY, NY 11530

EXAMINER

PHAM, HOAI V

ART UNIT

PAPER NUMBER

2892

DATE MAILED: 05/11/2015

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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14/177,481

02/11/2014

Nicolas Breil

FIS920130015US1(29690)

5539

TITLE OF INVENTION: SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	08/11/2015

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

45600 7590 05/11/2015
SCULLY, SCOTT, MURPHY & PRESSER, P.C.
400 GARDEN CITY PLAZA
Suite 300
GARDEN CITY, NY 11530

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/177,481	02/11/2014	Nicolas Breil	FIS920130015US1(29690)	5539

TITLE OF INVENTION: SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	08/11/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
PHAM, HOAI V	2892	438-572000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/177,481	02/11/2014	Nicolas Breil	FIS920130015US1(29690)	5539

45600 7590 05/11/2015
SCULLY, SCOTT, MURPHY & PRESSER, P.C.
400 GARDEN CITY PLAZA
Suite 300
GARDEN CITY, NY 11530

EXAMINER

PHAM, HOAI V

ART UNIT	PAPER NUMBER
----------	--------------

2892

DATE MAILED: 05/11/2015

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/177,481	Applicant(s) BREIL ET AL.	
	Examiner HOAI V. PHAM	Art Unit 2892	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 04/15/2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-9. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>2/11/2014</u> | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/HOAI V PHAM/
Primary Examiner, Art Unit 2892

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Election/Restrictions

1. Applicant's election without traverse of Species 2, claims 1-6 and 8-9 in the reply filed on 04/15/2015 is acknowledged.

Claim 7 is withdrawn from election/restriction since it depended on the allow claim 1.

Examiner's amendment

2. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

This application is in condition for allowance except for the presence of claims 10-20 non-elected without traverse. Accordingly, claims 10-20 have been cancelled.

3. The application has been amended as follows:

In the claims:

Cancel claims 10-20.

Allowable Subject Matter

4. Claims 1-9 are allowed.

5. The following is an examiner's statement of reasons for allowance: the prior art of record fails to show a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition-metal metal semiconductor alloy portion located on one side of said functional gate structure and located atop said source region; and a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of said functional gate structure and located atop said drain region while having the characteristics as recited in claim 1.

6. Any inquiry concerning this communication or earlier communications from the examiner should be directed to HOAI V. PHAM whose telephone number is (571)272-1715. The examiner can normally be reached on M-F.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Thao Xuan Le can be reached on 571-272-1708. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2892

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/HOAI V PHAM/

Primary Examiner, Art Unit 2892

Notice of References Cited	Application/Control No. 14/177,481	Applicant(s)/Patent Under Reexamination BREIL ET AL.	
	Examiner HOAI V. PHAM	Art Unit 2892	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-2005/0239287	10-2005	Wang et al.	438/682
	B	US-			
	C	US-			
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Receipt date: 02/11/2014

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

14177481 - GAI: 2892

Approved for use through 07/31/2012. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Nicolas Breil	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	FIS920130015US1(29690)	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	6063681		2000-05-16	Son	
	2	7737032		2010-06-15	Chan et al.	
	3	6815235		2004-11-09	Markle	
	4	7023055		2006-04-04	leong et al.	
	5	7329923		2008-02-12	Doris et al.	
If you wish to add additional U.S. Patent citation information please click the Add button.						Add
U.S.PATENT APPLICATION PUBLICATIONS						Remove
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20110287318		2011-11-24	Loveness et al.	

Receipt date: 02/11/2014 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14177481 - GAU: 2892	
	Filing Date			
	First Named Inventor	Nicolas Breil		
	Art Unit			
	Examiner Name			
	Attorney Docket Number	FIS920130015US1(29690)		

	2	20120175682		2012-07-12	Hagleitner et al.	
	3	20120112292		2012-05-10	Lavoie et al.	
	4	20100155954		2010-06-24	Mukherjee et al.	
	5	20070284693		2007-12-13	Kim et al.	
	6	20050116290		2005-06-02	de Souza et al.	

If you wish to add additional U.S. Published Application citation information please click the Add button. [Add](#)

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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² i	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							☐

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵	
	1		☐	

If you wish to add additional non-patent literature document citation information please click the Add button. [Add](#)

Receipt date: 02/11/2014 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14177481 - GAU: 2892	
	Filing Date			
	First Named Inventor	Nicolas Breil		
	Art Unit			
	Examiner Name			
	Attorney Docket Number	FIS920130015US1(29690)		

EXAMINER SIGNATURE			
Examiner Signature	/Hoai Pham/	Date Considered	04/29/2015
*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.			
¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.			

EAST Search History

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S87	375666	gate.clm.	US-PGPUB; USPAT; UPAD	OR	ON	2015/04/29 14:23
S88	192585	source.clm. and S87	US-PGPUB; USPAT; UPAD	OR	ON	2015/04/29 14:24
S89	141062	drain.clm. and S88	US-PGPUB; USPAT; UPAD	OR	ON	2015/04/29 14:24
S90	81122	semiconductor.clm. and S89	US-PGPUB; USPAT; UPAD	OR	ON	2015/04/29 14:24
S91	57426	substrate.clm. and S90	US-PGPUB; USPAT; UPAD	OR	ON	2015/04/29 14:25
S92	323	semiconductor adj alloy.clm. and S91	US-PGPUB; USPAT; UPAD	OR	ON	2015/04/29 14:25
S93	269	metal.clm. and S92	US-PGPUB; USPAT; UPAD	OR	ON	2015/04/29 14:26
S94	7	transition.clm. and S93	US-PGPUB; USPAT; UPAD	OR	ON	2015/04/29 14:26

4/ 30/ 2015 12:41:08 AM

<i>Index of Claims</i> 	Application/Control No. 14177481	Applicant(s)/Patent Under Reexamination BREIL ET AL.
	Examiner HOAI V PHAM	Art Unit 2892

✓	Rejected
=	Allowed

-	Cancelled
÷	Restricted

N	Non-Elected
I	Interference

A	Appeal
O	Objected

☐ Claims renumbered in the same order as presented by applicant										☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE													
Final	Original	03/16/2015	04/29/2015												
1	1	÷	=												
2	2	÷	=												
3	3	÷	=												
4	4	÷	=												
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Issue Classification 	Application/Control No. 14177481	Applicant(s)/Patent Under Reexamination BREIL ET AL.
	Examiner HOAI V PHAM	Art Unit 2892

CPC						
Symbol					Type	Version
H01L		29		456	F	2013-01-01
H01L		29		78	I	2013-01-01
H01L		29		16	I	2013-01-01
H01L		29		66545	I	2013-01-01
H01L		29		20	I	2013-01-01
H01L		29		66522	I	2013-01-01
H01L		21		28518	I	2013-01-01
H01L		21		2855	I	2013-01-01
H01L		29		452	I	2013-01-01
H01L		29		665	I	2013-01-01
H01L		21		28568	I	2013-01-01
H01L		21		28575	I	2013-01-01

CPC Combination Sets					
Symbol		Type	Set	Ranking	Version

NONE		Total Claims Allowed:	
(Assistant Examiner)		9	
(Date)			
/HOAI V PHAM/			
Primary Examiner.Art Unit 2892		04/29/2015	
(Primary Examiner)		(Date)	
		O.G. Print Claim(s)	O.G. Print Figure
		1	17

Issue Classification 	Application/Control No. 14177481	Applicant(s)/Patent Under Reexamination BREIL ET AL.
	Examiner HOAI V PHAM	Art Unit 2892

☐ Claims renumbered in the same order as presented by applicant															
☐ CPA															
☐ T.D.															
☐ R.1.47															
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6	6														
8	7														
7	8														
9	9														

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	9	
/HOAI V PHAM/ Primary Examiner.Art Unit 2892	04/29/2015	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	17

Search Notes 	Application/Control No. 14177481	Applicant(s)/Patent Under Reexamination BREIL ET AL.
	Examiner HOAI V PHAM	Art Unit 2892

CPC- SEARCHED		
Symbol	Date	Examiner
H01L 21/28518;H01L 21/8238;H01L 21/823814;H01L 27/092;H01L 29/49;H01L 29/4933;H01L 29/665;H01L 29/94	04/28/2015	HP

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
	INTERFERENCE SEARCHED PRINTOUT HISTORY	04/29/2015	HP

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 Alexandria, Virginia 22313-1450
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BIB DATA SHEET

CONFIRMATION NO. 5539

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
14/177,481	02/11/2014	438	2892	FS920130015US1(29690)
RULE				
APPLICANTS International Business Machines Corporation, Armonk, NY, Assignee (with 37 CFR 1.172 Interest);				
INVENTORS Nicolas Breil, Grenoble, FRANCE; Christian Lavoie, Ossining, NY; Ahmet S. Ozcan, Pleasantville, NY; Kathryn T. Schonenberg, Wappingers Falls, NY; Jian Yu, Danbury, CT;				
** CONTINUING DATA *****				
** FOREIGN APPLICATIONS *****				
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 02/24/2014				
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /HOAI V PHAM/ Acknowledged Examiner's Signature	☒ Met after Allowance HP Initials	STATE OR COUNTRY FRANCE	SHEETS DRAWINGS 12	TOTAL CLAIMS 20
INDEPENDENT CLAIMS 3				
ADDRESS SCULLY, SCOTT, MURPHY & PRESSER, P.C. 400 GARDEN CITY PLAZA Suite 300 GARDEN CITY, NY 11530 UNITED STATES				
TITLE SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS				
FILING FEE RECEIVED 1600	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit	

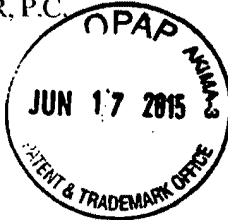
PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

45600 7590 05/11/2015
SCULLY, SCOTT, MURPHY & PRESSER, P.C.
400 GARDEN CITY PLAZA
Suite 300
GARDEN CITY, NY 11530



Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/177,481	02/11/2014	Nicolas Breil	FIS920130015US1(29690)	5539

TITLE OF INVENTION: SELF-ALIGNED LINER FORMED ON METAL SEMICONDUCTOR ALLOY CONTACTS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	08/11/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
PHAM, HOAI V	2892	438-572000

06/18/2015 SMOHAMM1 00000047 090458 14177481
01 FC:1501 960.00 DA

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☒ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.	Scully, Scott, Murphy & Presser, P.C. Catherine Ivers, Esq.
---	--	---

3. ASSIGNER NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE International Business Machines Corporation
(B) RESIDENCE: (CITY and STATE OR COUNTRY) Armonk, New York

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted: ☒ Issue Fee ☐ Publication Fee (No small entity discount permitted) ☐ Advance Order - # of Copies _____	4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above) ☐ A check is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☒ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number 09-0458/IBM (enclose an extra copy of this form).
--	--

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature Leslie S. Szivos
Typed or printed name Leslie S. Szivos, Ph.D.

Date 06-11-2015
Registration No. 39,394



PTO/SB/122 (11-08)
Approved for use through 11/30/2011. OMB 0651-0035
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CHANGE OF CORRESPONDENCE ADDRESS Application

Address to:
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Application Number	14177481
Filing Date	02-11-2014
First Named Inventor	Nicolas Breil
Art Unit	2892
Examiner Name	Hoai V. Pham
Attorney Docket Number	FIS920130015US1(29690)

Please change the Correspondence Address for the above-identified patent application to:

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Customer Number:

32074

OR

☐ Firm or
Individual Name

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State

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I am the:

☐ Applicant/Inventor

☐ Assignee of record of the entire interest.
Statement under 37 CFR 3.73(b) is enclosed. (Form PTO/SB/96).

☒ Attorney or agent of record. Registration Number 39,394

☐ Registered practitioner named in the application transmittal letter in an application without an executed oath or declaration. See 37 CFR 1.33(a)(1). Registration Number _____

Signature

Typed or Printed
Name

Leslie S. Szivos, Ph.D.

Date 06-11-2015

Telephone (516) 742-4343

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.

☒ *Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/177,481	07/28/2015	9093425	FIS920130015US1(29690)	5539

32074 7590 07/08/2015
INTERNATIONAL BUSINESS MACHINES CORPORATION
DEPT. 18G
BLDG. 321-482
2070 ROUTE 52
HOPEWELL JUNCTION, NY 12533

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

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